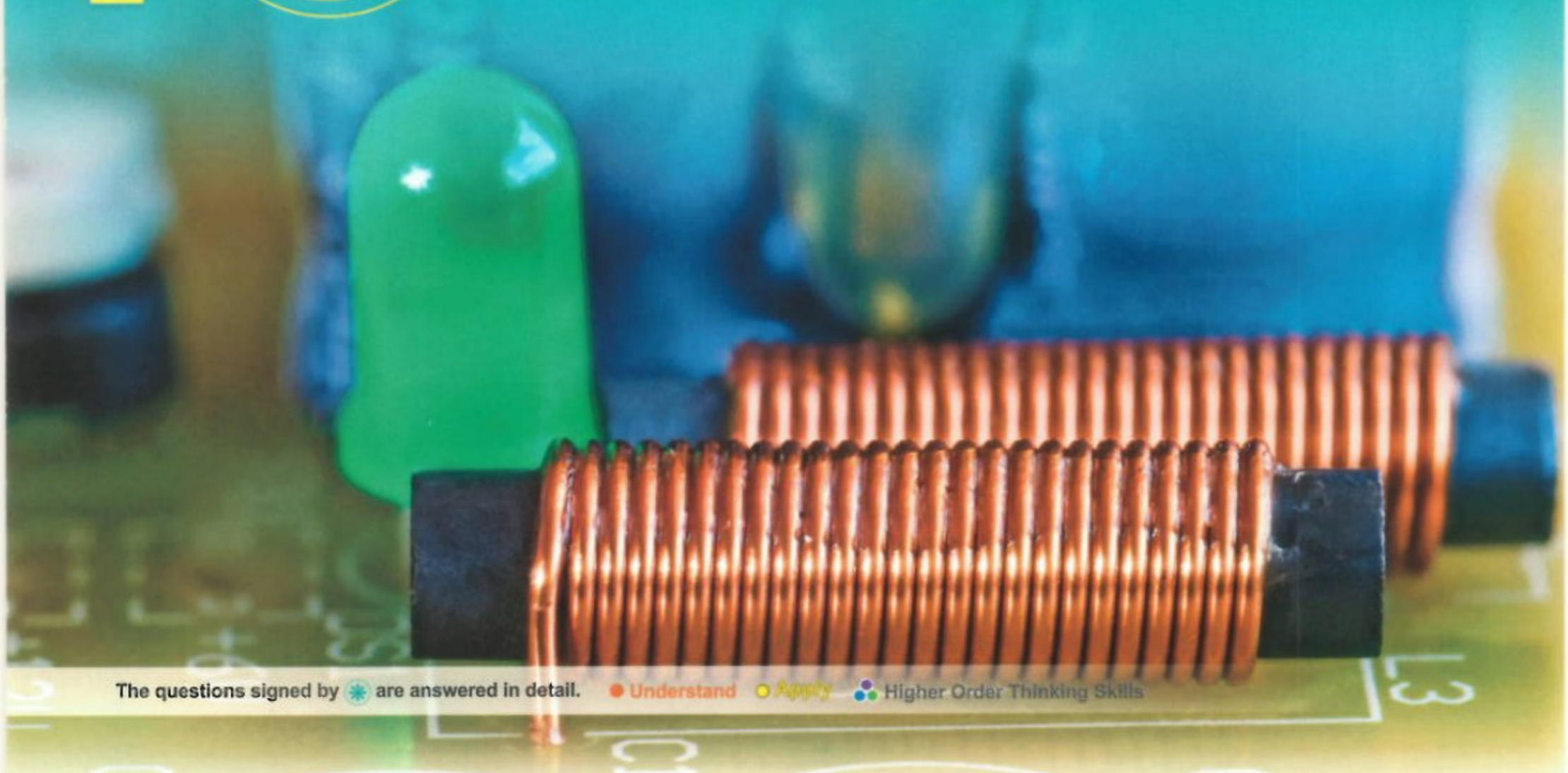
Chapter 2

Questions on Lesson Two

Following of the Magnetic Effect of the Electric Current

Interactive test



The questions signed by ✱ are answered in detail. ● Understand ● Apply Higher Order Thinking Skills

Important guidelines

- **The magnetic flux density (B) at the center of a circular coil:** $B = \frac{\mu N I}{2 r}$

- **The number of turns in a circular coil:**

  If a wire of length ℓ is coiled to make a coil of radius r:

  $$N = \frac{\ell \text{ (length of the coil's wire)}}{2 \pi r \text{ (circumference of the turn)}}$$

  Where (N is either an integer or a rational number).

  If the coil is a part of a circle (arc) as shown in the figure:

  $$N = \frac{\theta}{360}$$



- **In the case of reshaping a circular coil where the number of turns is changed from N_1 to N_2 and connecting it to the same voltage:**

$$\because \frac{B_1}{B_2} = \frac{N_1 r_2}{N_2 r_1} \quad , \quad \frac{N_1}{N_2} = \frac{r_2}{r_1} \quad \therefore \frac{B_1}{B_2} = \frac{N_1^2}{N_2^2} = \frac{r_2^2}{r_1^2}$$

CamScanner

Scanned with OKEN Scanner

CamScanner

The net magnetic flux density at the common center of two coils

In one plane

Two currents are in the same direction

$B_t = B_1 + B_2$



Two currents are in opposite directions

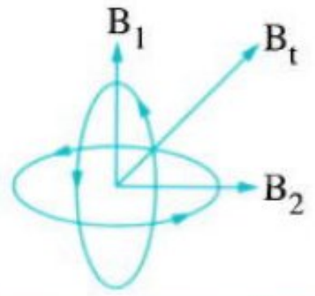
$B_t = B_1 - B_2$

$(B_1 > B_2)$



Perpendicular

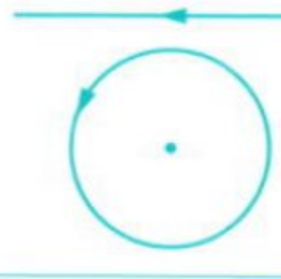
$B_t = \sqrt{B_1^2 + B_2^2}$



The net magnetic flux density in the case of a circular coil and a wire in the same plane carrying electric currents if the field of each of the wire and the coil are:

In the same direction

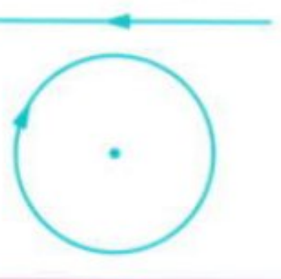
$B_t = B_{coil} + B_{wire}$



In two opposite directions

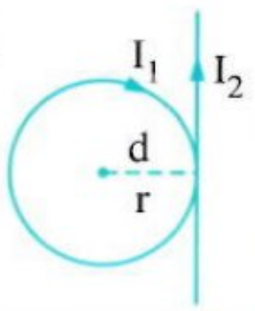
$B_t = B_{coil} - B_{wire} \quad (B_{coil} > B_{wire})$

$B_t = B_{wire} - B_{coil} \quad (B_{wire} > B_{coil})$



- **In the case of a circular coil tangent to a straight wire where the total magnetic flux at the center vanishes:** $B_{coil} = B_{wire}$

$$\frac{\mu I_1 N}{2r} = \frac{\mu I_2}{2\pi d} \qquad NI_1 = \frac{I_2}{\pi}$$



- **The magnetic flux density (B) at the center of a solenoid:** $B = \mu \frac{N I}{\ell} = \mu n I$

When the turns are tangent to each other along the coil, the length of the coil: $\ell = N \times 2\grave{r}$

(Where: n is the number of turns per unit length, $\grave{r}$ is the radius of the wire)

Scanned with OKEN Scanner

- **When the turns of the circular coil are displaced uniformly away from each other, so it becomes a solenoid having the same number of turns and carrying the same electric current as the circular coil, we can compare between them according to the following relation:**

$$\frac{B_{circular}}{B_{solenoid}} = \frac{\ell_{solenoid}}{2\,r_{circular}}$$

The net flux density at the common center of two solenoid coils when the two currents are:

In the same direction	In two opposite directions
$B_t = B_1 + B_2$	$B_t = B_1 - B_2$ $(B_1 > B_2)$

- **If a wire is placed parallel or perpendicular to the axis of a solenoid and electric currents pass through each of them (the two fields are perpendicular), then net magnetic flux density at a point on the solenoid axis and at a certain distance from the straight wire is:**

$$B_t = \sqrt{B_{wire}^2 + B_{coil}^2}$$

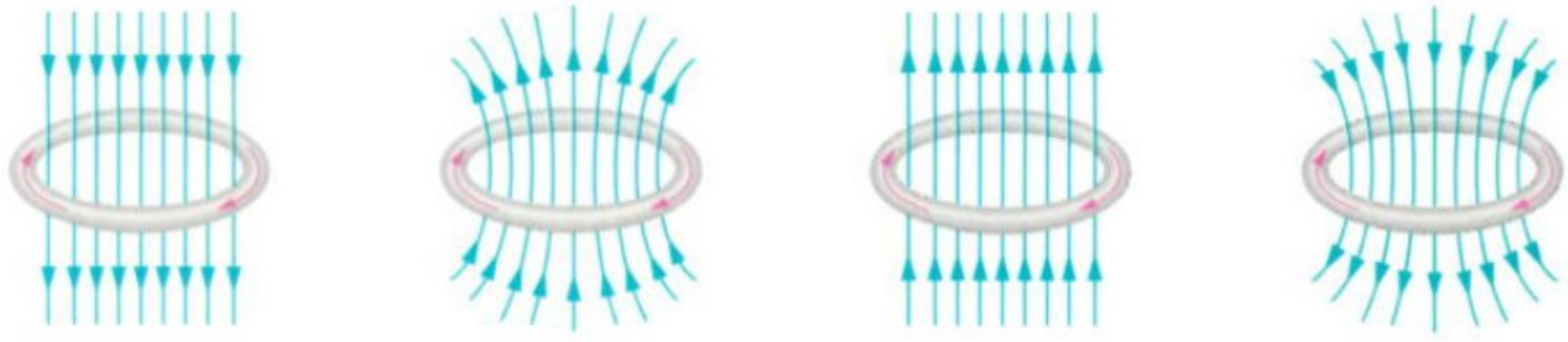
- **If a wire is put perpendicular to the axis of a solenoid but away from it by a distance d, the formed magnetic fluxes due to the two currents that pass in the wire and the solenoid are added or subtracted with applying Ampere's right hand rule.**

First Multiple choice questions

Use the following constants when needed: $\mu_{air} = 4\pi \times 10^{-7}$ Wb/A.m, $e = 1.6 \times 10^{-19}$ C

Circular coils

1 Which of the following figures correctly represents the magnetic flux lines pattern produced due to the flow of a direct electric current through a circular metallic ring?



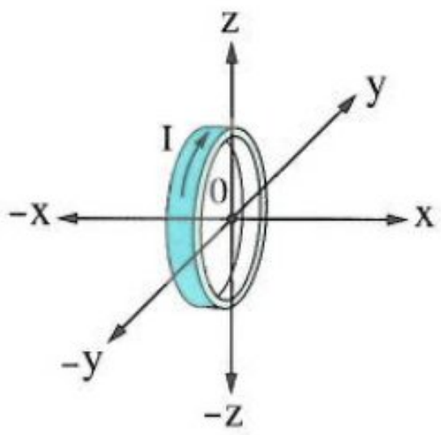
a b c d

CamScanner

Scanned with OKEN Scanner

CamScanner

2 A circular metallic ring carries an electric current as shown in the opposite figure, which of the following directions represents the direction of the magnetic field at the center of the ring that is formed due to the flow of the current in the ring?



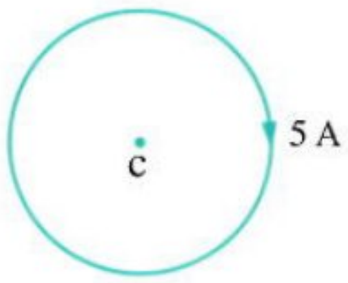
(a) The positive direction of x-axis.

(b) The positive direction of z-axis.

(c) The negative direction of x-axis.

(d) The negative direction of y-axis.

3 In the opposite figure, a circular ring of radius 10 cm carries an electric current of intensity 5 A, then the magnetic flux density at the center of the ring (c) and its direction are ..... .



(Where: $\pi = 3.14$)

(a) 2.6×10^{-5} T, perpendicular out of the page

(b) 3.14×10^{-5} T, perpendicular out of the page

(c) 3.14×10^{-5} T, perpendicular into the page

(d) 2.6×10^{-5} T, perpendicular into the page

4 If the magnetic flux density at the center of a circular ring of radius 4π cm is 5×10^{-5} T and the magnetic permeability of air is $4\pi \times 10^{-7}$ Wb/A.m, so the electric current intensity through the ring is ..... .

(a) 7 A (b) 7.14 A (c) 10 A (d) 17 A

5 A circular coil carries an electric current that produces a magnetic field of density B at the center of the coil. When the electric current intensity is increased to the double and the diameter of the coil is increased to the double with keeping the same number of turns, then the magnetic flux density at the center of the coil equals ..... .

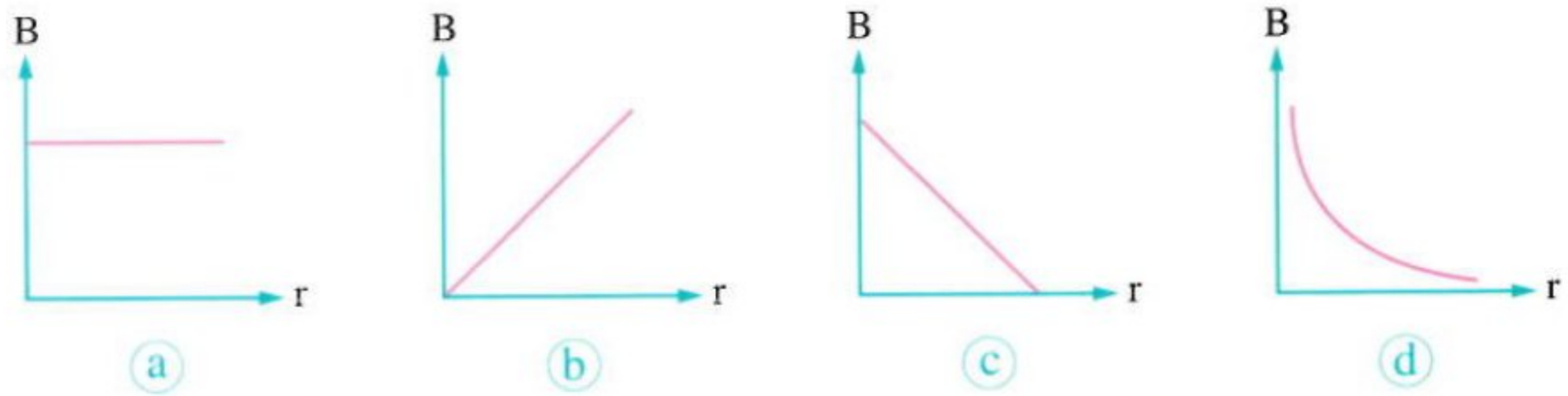
(a) B (b) 2 B (c) $\frac{B}{2}$ (d) $\frac{B}{4}$

6 Two metallic rings (x and y) carry electric currents of intensities I and 2I respectively such that the magnetic flux densities at their centers are equal. The ratio between the diameters of the two rings $\left(\frac{d_x}{d_y}\right)$ equals ..... .

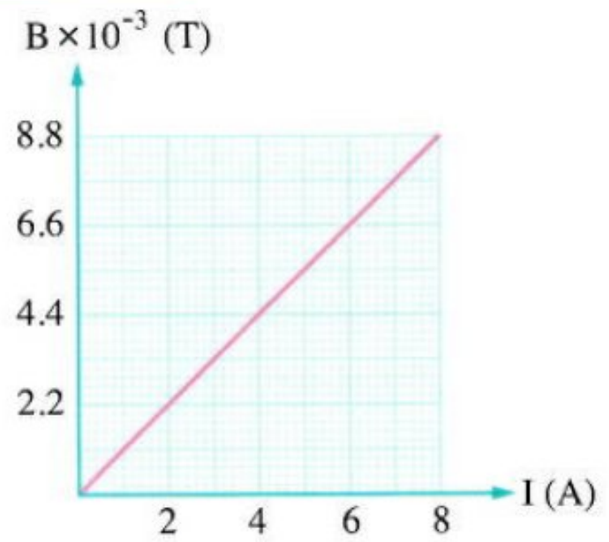
(a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{1}{4}$ (d) $\frac{4}{1}$

Scanned with OKEN Scanner

7 Multiple circular coils have the same number of turns and carry the same electric current, so which of the following graphs represents the relation between the magnitude of the magnetic flux density (B) at the center of each coil and the radius (r) of the coil?

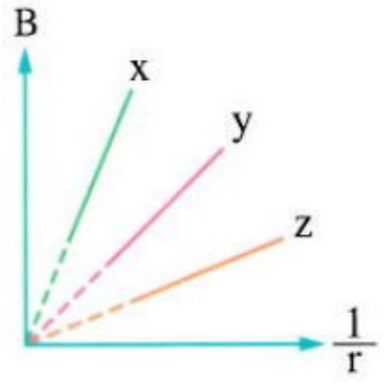


8 The opposite graph represents the relation between the magnetic flux density (B) at the center of a circular coil that consists of 350 turns and the current intensity (I) in the coil, so the diameter of the coil equals ..... .



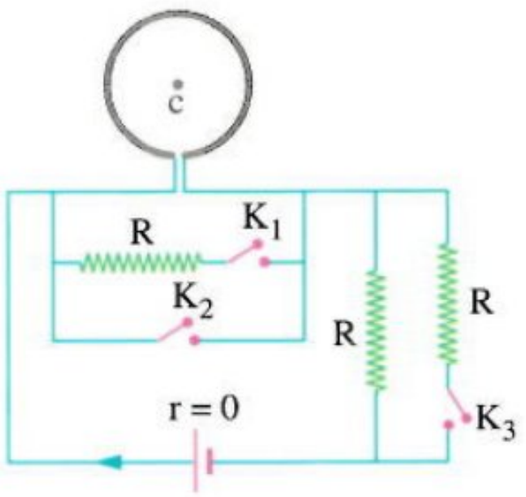
(a) 10 cm (b) 20 cm
(c) 30 cm (d) 40 cm

9 Three groups of metallic rings (x, y and z) carry electric currents with intensities I_x, I_y and I_z respectively. The opposite graph represents the relation between the magnetic flux density (B) at the center of the ring and the reciprocal of its radius $\left(\frac{1}{r}\right)$. The correct order of the current intensities is ..... .



(a) $I_x < I_y < I_z$ (b) $I_x > I_y > I_z$ (c) $I_x > I_z > I_y$ (d) $I_x = I_y = I_z$

10 A wire of resistance R is wound as a circular coil of center c and it is connected to an electric circuit as shown in the opposite diagram. Which switches in the circuit cause an increase in the magnetic flux density at the center of the coil when closed, as a result of the current passing through it?



(a) Switch K_1 only. (b) Switch K_2 only.
(c) Switch K_3 only. (d) The three switches.

11 ✳ A circular coil of radius 10 cm was made of a wire of resistivity 10^{-6} Ω.m and cross-sectional area 0.4×10^{-4} m^2, if the coil is connected to an electric source of emf V_B whose internal resistance is negligible, the magnitude of the magnetic flux density at its center becomes 0.01 T. So, the emf of the electric source equals ..... .

(a) 10 V (b) 15 V (c) 20 V (d) 25 V

CamScanner
Scanned with OKEN Scanner

CamScanner

12 A circular coil of radius r is connected across a voltage V, then the magnetic flux density at the center of the coil was B. When half of the coil's turns are remov ed, and the remaining part of the coil is connected to the same voltage V, the magnetic flux density at the coil's center ..... .

(a) remains B (b) becomes 2 B (c) becomes $\frac{B}{4}$ (d) becomes $\frac{B}{2}$

13 * A copper wire of length 50.24 m and cross-sectional area 1.79×10^{-7} m^2 has been coiled in the form of a circular coil of radius 4 cm, then its terminals were connected to a direct current source of emf 12 V and internal resistance 1 Ω. Keeping in mind that the resistivity of copper is 1.79×10^{-8} Ω.m, so the magnetic flux density at the center of the coil equals ..... .

(Take: $\pi = 3.14$)

(a) 3.125×10^{-4} T (b) 5.93×10^{-4} T (c) 6.25×10^{-3} T (d) 8.12×10^{-3} T

14 * A wire of length ℓ is shaped into a circular coil with area A. When an electric current of intensity I passes through the coil, the magnetic flux density (B) at the center of the coil can be determined from the relation ; .....

(a) $B = \frac{\mu I \ell}{A}$ (b) $B = 4 \times \frac{\mu I \ell}{A}$ (c) $B = \frac{1}{2} \times \frac{\mu I \ell}{A}$ (d) $B = \frac{1}{4} \times \frac{\mu I \ell}{A}$

15 * A straight wire of length 31.4 m is coiled in the form of a circular coil of face area 3.14 cm^2 and an electric current of 1 A is set up through it, then the magnetic flux density at the center of the coil equals ..... .

(Take: $\pi = 3.14$)

(a) 1.57×10^{-2} T (b) 3.14×10^{-2} T (c) 6.28×10^{-2} T (d) 9.42×10^{-2} T

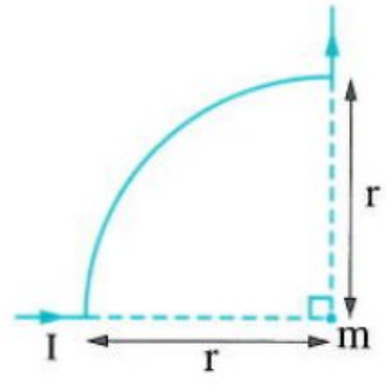
16 A long straight wire carries an electric current of intensity 5 A, so at a point at a perpendicular distance 5 cm away from mid-point of the wire's axis, the magnetic flux density was B. If a part of length 120 cm is cut from this wire and wound into a circular coil with a radius 9.55 cm where the same current passed through it, the magnetic flux density at the center of the coil ..... .

(Take: $\pi = 3.14$)

(a) exceeds B by 8.6×10^{-5} T (b) exceeds B by 4.6×10^{-5} T

(c) becomes less than B by 8.6×10^{-5} T (d) becomes less than B by 4.6×10^{-5} T

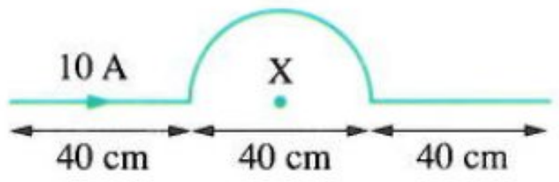
17 In the opposite figure, a part of a long straight wire is bent to form a quarter of a circle, then a current of intensity I is set up through it, so the magnetic flux density at point m equals ..... .



(a) $\frac{\mu I}{2r}$ (b) $\frac{\mu I}{4r}$ (c) $\frac{\mu I}{6r}$ (d) $\frac{\mu I}{8r}$

Scanned with OKEN Scanner

18 ✳ In the opposite figure, a part of a long wire is bent to form half a circle of diameter 40 cm, then a current of intensity 10 A is set up through the wire, the magnetic flux density at point X equals ..... .



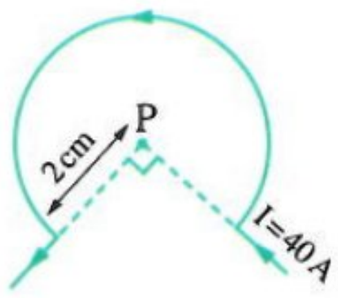
(a) 3.14×10^{-5} T (b) 6.28×10^{-5} T (c) 1.57×10^{-5} T (d) 2.826×10^{-4} T

19 ✳ In the opposite figure:

(i) The magnetic flux density at point P equals ..... .

(a) 9.43×10^{-5} T (b) 6.28×10^{-5} T

(c) 3.14×10^{-5} T (d) 9.43×10^{-4} T



(ii) The direction of the magnetic flux at point P is ..... .

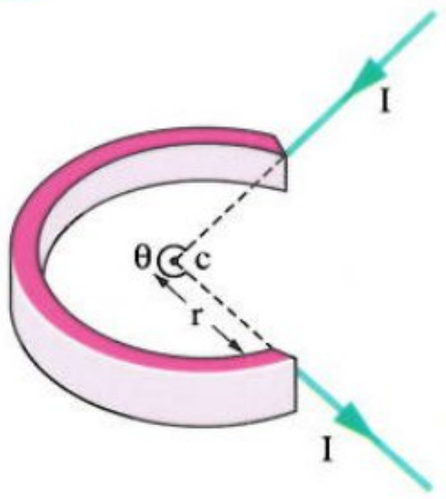
(a) perpendicular into the page

(b) perpendicular out of the page

(c) parallel to the plane of the page and rightwards

(d) parallel to the plane of the page and leftwards

20 The opposite figure represents a part of a circular metallic ring of radius (r) placed on a horizontal surface and carries an electric current of intensity I, if the angle (θ) was measured in degrees, then the magnetic flux density at center (c) is determined from the relation; .....



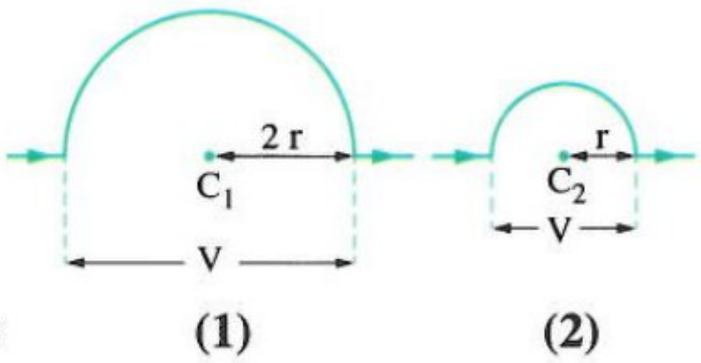
(a) $B = \frac{\mu I \theta}{r}$ (b) $B = \frac{\mu I \theta}{360\ r}$

(c) $B = \frac{\mu I \theta}{2 \pi r}$ (d) $B = \frac{\mu I \theta}{720\ r}$

21 ✳ If an electric current passes through a wire of length 26.4 cm which is curved as an arc of a circle of radius 5.6 cm, a magnetic flux of density 8.25×10^{-6} T at the center of this circle is formed, so the intensity of the passing electric current equals ..... .

(a) 1.5 A (b) 0.98 A (c) 0.75 A (d) 0.49 A

22 ✳ In the opposite figure, two metallic wires are bent to form two half rings of different radii, the two wires have the same cross-sectional areas and are made of a material of large resistivity, when the same potential difference is applied across each of them, the magnetic flux density at C_1 equals B, hence at C_2, it equals ..... .



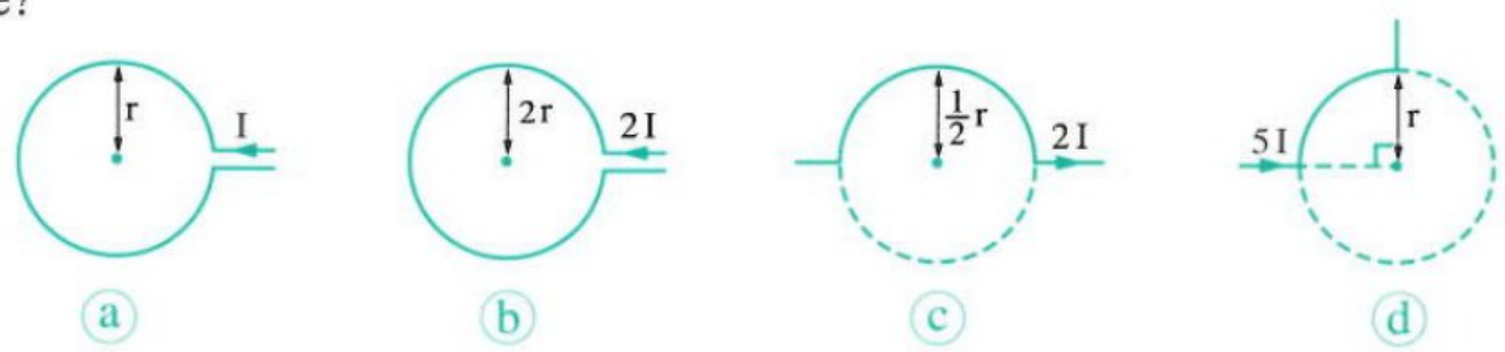
(a) $\frac{B}{2}$ (b) 2 B (c) 3 B (d) 4 B

CamScanner

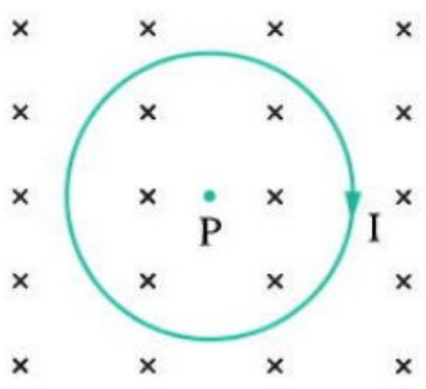
Scanned with OKEN Scanner

CamScanner

23 ✱ In which of the following coils, the magnetic flux density at its center has the largest value?

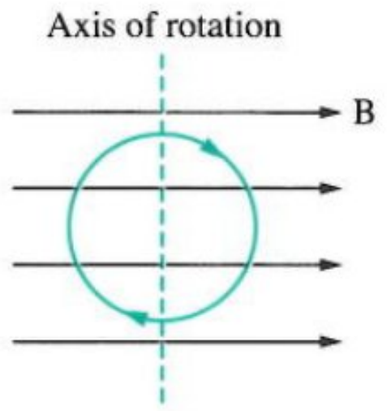


24 The opposite figure represents a circular coil of radius 8 cm that consists of 14 turns which is placed in the plane of the page and carries an electric current of intensity 12 A. If an external uniform magnetic field of flux density 10^{-5} T is directed perpendicular into the plane of the coil, so the magnitude and the direction of the net magnetic flux density at the center of the coil (P) are ..... , ...



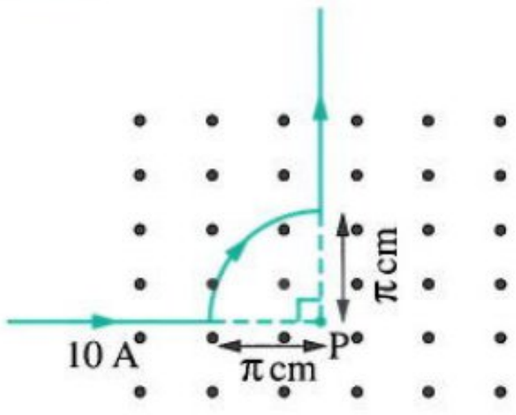
	The magnitude of the net magnetic flux density at the center of the coil	The direction of the net magnetic flux density at the center of the coil
a	1.33×10^{-3} T	perpendicular into the page
b	1.33×10^{-3} T	perpendicular out of the page
c	3.13×10^{-3} T	perpendicular into the page
d	3.13×10^{-3} T	perpendicular out of the page

25 In the opposite figure, when a current-carrying circular coil was placed parallel to a uniform magnetic field of flux density B, the resultant magnetic flux density at the center of the coil was $\sqrt{5}$ B, so when the coil rotates 90°, the resultant magnetic flux density at the center of the coil could be ..... .



(a) B or 2 B (b) B or 5 B
(c) B or 3 B (d) zero or B

26 ✱ In the opposite figure, a part of a straight wire is reshaped in the form of quarter of a circle and placed in the plane of the page. If an external magnetic field of flux density 6×10^{-6} T that is directed perpendicular out of the page affects the wire, then the net magnetic flux density at its center P equals ..... .



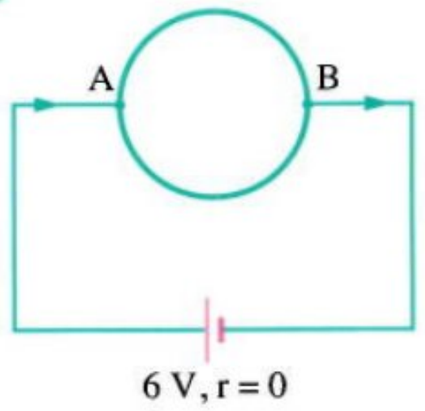
(a) 11×10^{-5} T (b) 5.6×10^{-5} T (c) 4.4×10^{-5} T (d) 0

Scanned with OKEN Scanner

27 ✳ A circular ring of radius 5 cm carries an electric current of 10 A. If the ring is bent such that each half ring becomes perpendicular to the other half, the net magnetic flux density at the center becomes ..... .

(a) 8.9×10^{-5} T (b) 7.3×10^{-5} T (c) 13.21×10^{-6} T (d) 12.5×10^{-6} T

28 ✳ A straight wire of resistance 48 Ω is wound as a closed ring of diameter 0.1 m, then a battery of 6 V whose internal resistance is negligible is connected across a diameter of the ring as in the opposite figure, so:



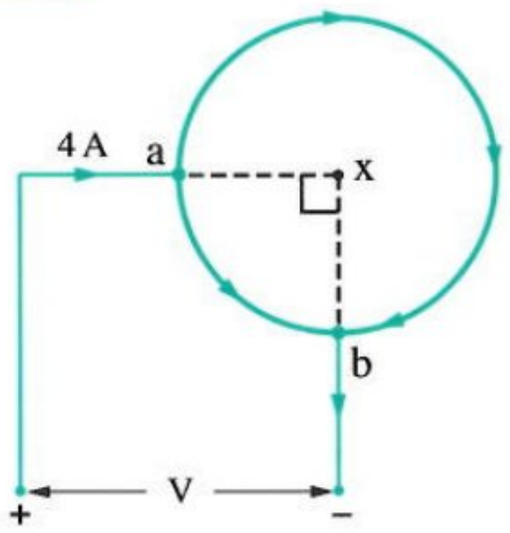
(i) The electric current intensity that passes through either half of the ring equals ..... .

(a) 0.75 A (b) 0.25 A (c) 0.36 A (d) 0.41 A

(ii) The magnetic flux density at the center of the ring equals ..... .

(a) 0 (b) $\pi \mu$ T (c) $2 \pi \mu$ T (d) $3 \pi \mu$ T

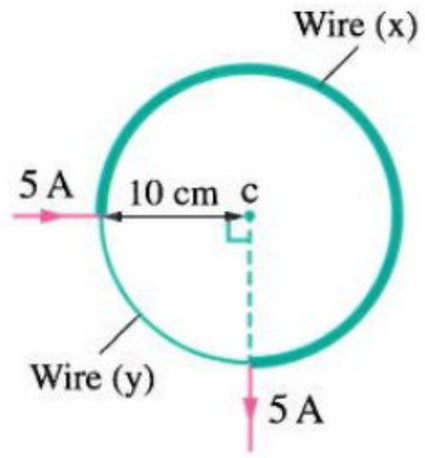
29 ✳ In the opposite figure, a metallic ring of center x is connected to an electric source via two points (a, b) as shown in the opposite figure, so that an electric current passes through the ring, hence the net magnetic flux density at the center of the ring equals ..... .



(a) $\frac{3 \mu}{8 r}$ (b) $\frac{6 \mu}{8 r}$

(c) $\frac{4 \mu}{8 r}$ (d) 0

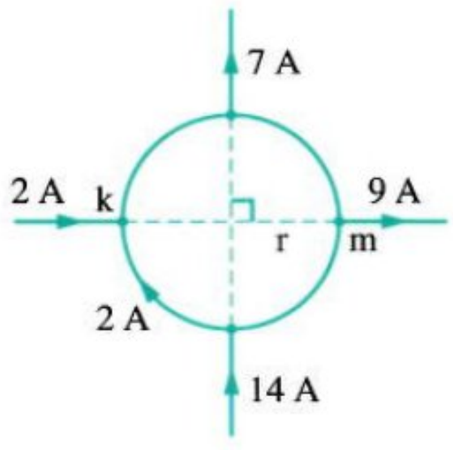
30 Two wires x, y of the same metal, the cross-sectional area of wire (x) is double that of wire (y), they are connected together to form a metallic circular ring and an electric current of intensity 5 A passed through it as shown in the opposite figure, then the magnetic flux density at the center (c) equals ..... .



(a) 1.26×10^{-5} T (b) 7.54×10^{-6} T

(c) 4.71×10^{-6} T (d) 3.77×10^{-6} T

31 An electric current is set up into a metallic circular ring through multiple paths as shown in the opposite figure, so the direction of the magnetic flux density at the center of the ring will be ..... .



(a) perpendicular into the plane of the ring

(b) perpendicular out of the plane of the ring

(c) rightward towards point m

(d) leftward towards point k

CamScanner

Scanned with OKEN Scanner

CamScanner

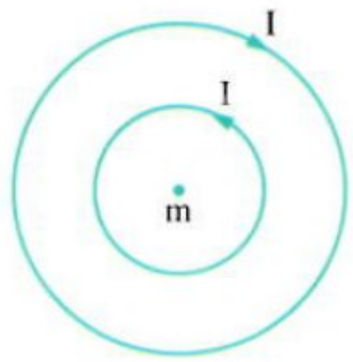
32 A coil that has one turn of wire is carrying an electric current of intensity I, then the same wire of that coil is recoiled as a circular coil of 4 turns while the intensity of the passing electric current becomes $\frac{I}{2}$, so the ratio between the magnetic flux density at the center of the coil in the first case to that in the second case equals ..... .

(a) $\frac{4}{9}$ (b) $\frac{9}{4}$ (c) $\frac{1}{8}$ (d) $\frac{8}{1}$

33 ✳ According to Bohr's model of the hydrogen atom, the electron rotates around the nucleus in a circular orbit of radius 5.3×10^{-11} m with a frequency of 6.6×10^{15} Hz, hence the magnetic flux density at the center of the orbit due to the rotation of the electron equals ..... .

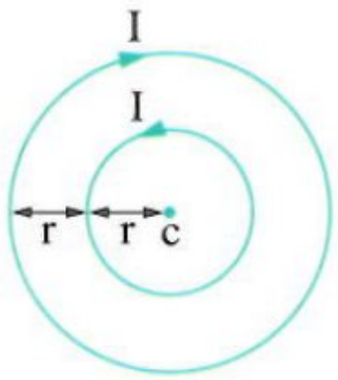
(a) 12.52 T (b) 8.48 T (c) 5.65 T (d) 4.24 T

34 Two concentric metallic rings are placed in the same plane, both carry an electric current of intensity I as in the figure. So the direction of the produced magnetic flux at the center point m is ..... .



(a) to the right (b) to the left
(c) into the page (d) out of the page

35 Two concentric metallic rings are placed in the same plane, both carry an electric current of the same intensity I and in opposite directions as shown in the opposite figure, which of the following changes would lead to a reversal in the direction of the resultant magnetic flux density at the center of the two rings (c) ?



(a) Reducing the current in the inner ring to $\frac{2}{3}$ I
(b) Reducing the current in the inner ring to $\frac{1}{2}$ I
(c) Increasing the current in the outer ring to 2 I
(d) Increasing the current in the outer ring to 3 I

36 ✳ Two concentric circular coils are lying in the same plane. The two coils have 400 turns and 500 turns, their radii are 20 cm and 10 cm respectively and they are carrying electric currents in the same direction of intensities 7 A and 10 A respectively, so:

(i) The magnetic flux density at the concentric center of the two coils equals ..... .

(a) 16.3×10^{-3} T (b) 32.6×10^{-3} T (c) 40.2×10^{-3} T (d) 81.5×10^{-3} T

(ii) The magnetic flux density at the concentric center of the two coils, when one of them is rotated about the concentric center:

① by an angle of 180° to be overturned, equals ..... .

(a) 22.6×10^{-3} T (b) 45.3×10^{-3} T (c) 67.8×10^{-3} T (d) 90.4×10^{-3} T

② by an angle of 90° to be perpendicular to the other coil, equals ..... .

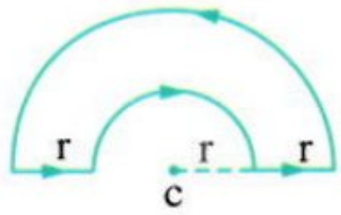
(a) 18.5×10^{-3} T (b) 32.6×10^{-3} T (c) 65.2×10^{-3} T (d) 97.8×10^{-3} T

Scanned with OKEN Scanner

37 ✳ Two concentric circular coils that are lying in the same plane carry two electric currents of equal intensities and opposite directions, if the diameter of the first is 10 cm and it has 100 turns and the diameter of the second is 20 cm, hence the number of turns of the second coil that makes the magnetic flux density at their center vanish, equals ..... .

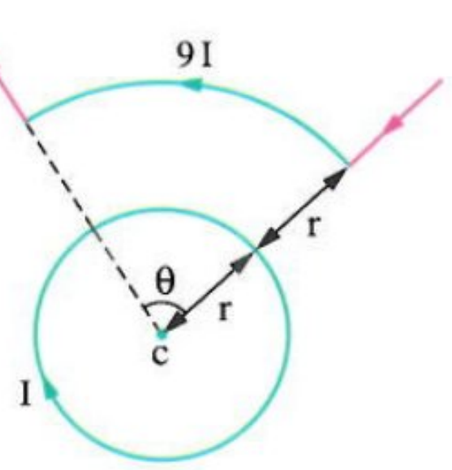
(a) 50 turns (b) 100 turns (c) 150 turns (d) 200 turns

38 In the opposite figure, if an electric current of intensity 1 A passes through the wires, then the resultant magnetic flux density that is produced at point c is ..... .



(a) $\frac{\mu}{5\,r}$ (b) $\frac{\mu}{2\,r}$ (c) $\frac{\mu}{4\,r}$ (d) $\frac{\mu}{8\,r}$

39 A circular ring and a part of another circular ring are lying concentrically in the same plane, the radii of them respectively are r, 2r when two electric currents of intensities I, 9I pass through them as shown in the opposite figure, if the resultant of the magnetic flux density at center (c) equals zero, then the angle (θ) is equal to ..... .

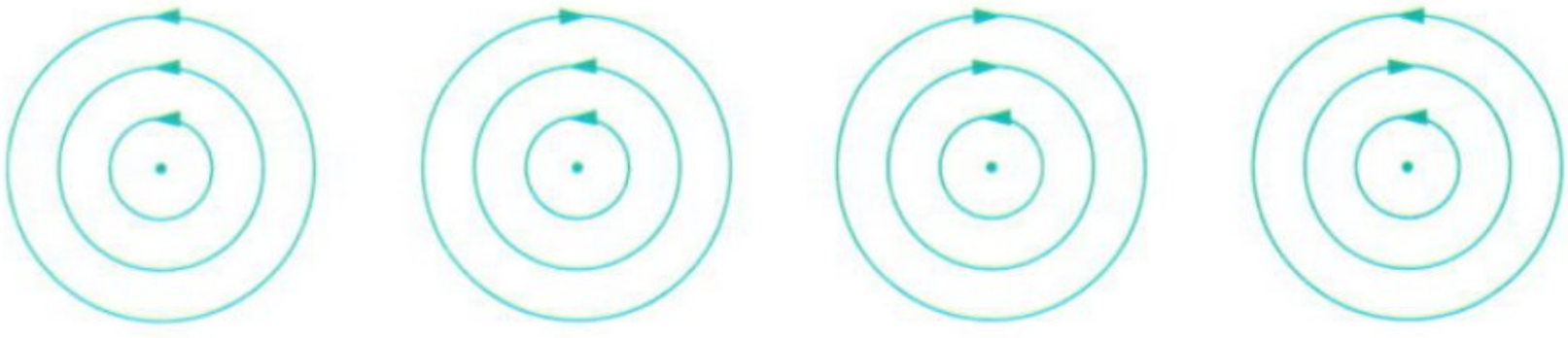


(a) 60° (b) 80°
(c) 90° (d) 95°

40 ✳ Two circular coils are lying concentrically in the same plane where the diameter of the first is double that of the second and both of them carry electric currents of the same intensity and in the same direction, hence at their center B_1 (of the external coil) < B_2 (of the internal coil) and when the current is reversed in the external coil, the magnetic flux density at the center decreases to its half, so the ratio between the number of turns of the two coils $\left(\frac{N_1}{N_2}\right)$ equals ..... .

(a) $\frac{4}{5}$ (b) $\frac{6}{11}$ (c) $\frac{2}{3}$ (d) $\frac{9}{8}$

41 In each of the following figures, three concentric metallic rings are in the same plane, carrying electric currents of the same intensity. In which of these figures is the magnetic flux density at the center the least?

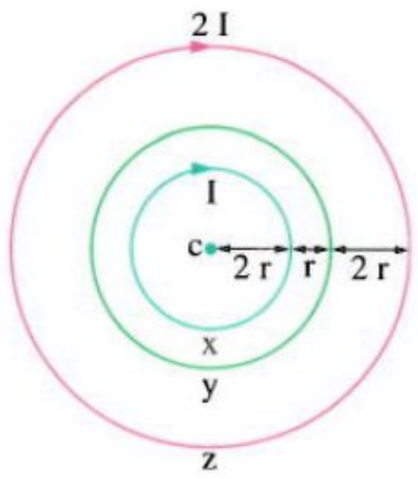


(a) (b) (c) (d)

CamScanner
Scanned with OKEN Scanner

CamScanner

42 ✱ Three concentric rings x, y and z of center c, all in the same plane and each of them carries an electric current as in the opposite figure. If the net magnetic flux density at the center (c) equals zero, then the intensity and the direction of the electric current that passes in the ring y are ..... .



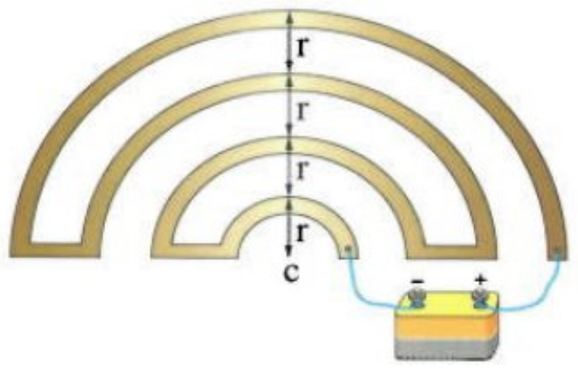
(a) 1.5 I, in the clockwise direction

(b) 1.5 I, in the anti-clockwise direction

(c) 2.7 I, in the clockwise direction

(d) 2.7 I, in the anti-clockwise direction

43 Four semi-circular metallic rings are connected together to a battery as shown in the opposite figure, then the direction of the resultant magnetic flux at the common center (c) is ..... .



(a) perpendicular to the page and outward

(b) perpendicular to the page and inward

(c) in the plane of the page and rightward

(d) in the plane of the page and leftward

44 A long straight wire carrying an electric current of intensity I generates a magnetic flux of density B at a point perpendicular to the wire at a distance d. If the same current (I) passes through a circular loop, producing the same magnetic flux density B at its center, the radius of the loop is ..... .

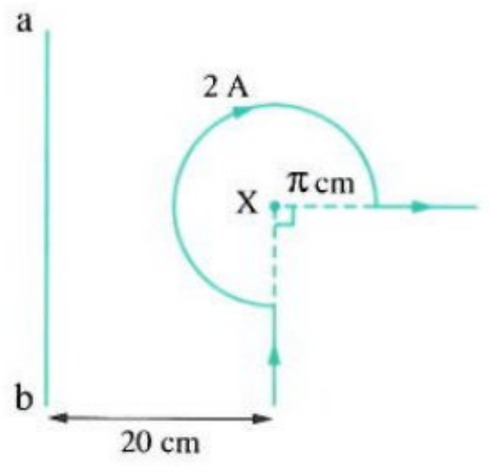
(a) greater than d

(b) less than d

(c) equal to d

(d) indeterminable

45 The opposite figure shows a very long straight wire that carries an electric current of intensity I. The wire is placed in the same plane with a part of a metallic ring and at a distance 20 cm from the center (X) of the ring. So, the magnitude and the direction of the current I that makes the magnetic flux density at the center of the ring equal to zero are ..... .



(a) 15 A, from a to b

(b) 30 A, from a to b

(c) 15 A, from b to a

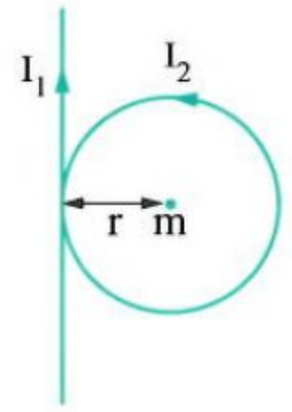
(d) 30 A, from b to a

Scanned with OKEN Scanner

46 ✳ A long straight wire is placed vertically tangent to a circular coil of one turn that is lying in the plane of the magnetic meridian of the Earth and carries a current of intensity 0.42 A. A free movable magnetic needle is placed in the horizontal plane at the center of the coil. So, the intensity of the electric current which when passes in the wire causes no deflection in the needle, equals ..... .

(a) 0.96 A (b) 1.07 A (c) 1.32 A (d) 2.56 A

47 In the opposite figure, a long straight wire is tangent to a circular coil where both of the wire and the coil are in the same plane and isolated from each other and both carry electric currents that flow as shown in the figure, so the net magnetic flux density at the center of the circular coil was equal to zero. If the coil is overturned, the net magnetic flux density at its center becomes ..... .



(a) $\frac{B_{coil}}{2}$ (b) B_{coil} (c) B_{wire} (d) $2\,B_{wire}$

48 An insulated straight wire carries an electric current of intensity I, if the wire is placed tangent to a semi-circular ring either parallel to its plane or perpendicular to it and it carries an electric current of intensity I in the shown direction of the following figures.



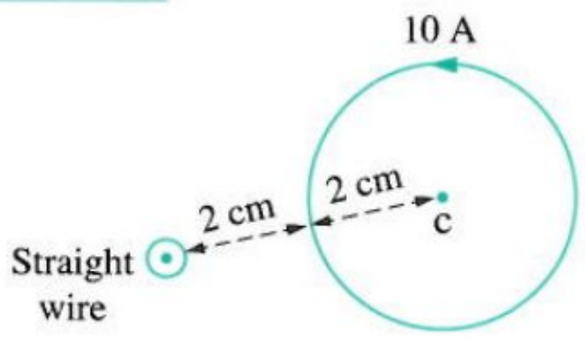
(1) (2) (3) (4)

So, the correct order for the values of the magnetic flux density at the center of the semi-circular ring at that cases is .....

(a) $B_4 > B_2 > B_3 > B_1$ (b) $B_1 > B_3 = B_4 > B_2$

(c) $B_2 > B_3 = B_4 > B_1$ (d) $B_3 = B_4 > B_2 > B_1$

49 The opposite figure represents a metallic ring, if a long straight wire is placed near to and perpendicular to its plane and an electric current of intensity 10 A passes through each of them. Then the resultant magnetic flux density at the center (c) of the ring is approximately equal to ..... .



(a) 10^{-4} T (b) 2.6×10^{-4} T (c) 3.2×10^{-4} T (d) 3.6×10^{-4} T

CamScanner
Scanned with OKEN Scanner

CamScanner

50 An insulated long wire carries an electric current of intensity I, a part of it was wrapped in the form of a circular ring of radius r by two ways as shown in the following two figures:

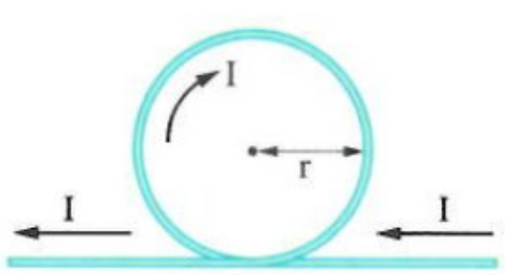


Figure (1)

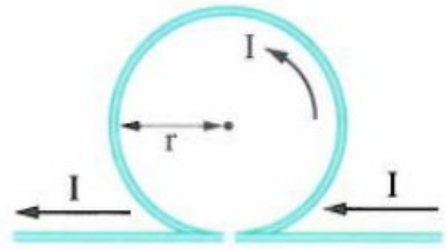
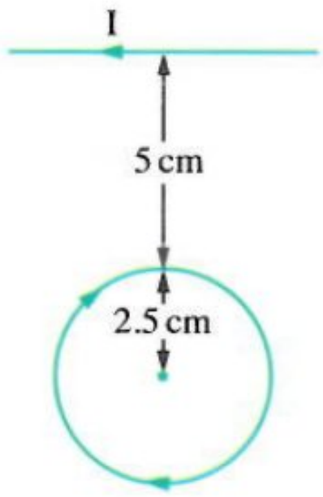


Figure (2)

Then, the ratio between the resultant of the magnetic flux density at the center of the ring in both cases $\left(\frac{B_1}{B_2}\right)$ is ..... .

(a) $\frac{1}{1}$ (b) $\frac{\pi + 1}{\pi}$ (c) $\frac{1}{\pi - 1}$ (d) $\frac{\pi + 1}{\pi - 1}$

51 ✳ A circular ring of radius 2.5 cm that carries an electric current of 3 A is placed at a distance of 5 cm from a straight wire that carries an electric current I as in the figure, so:



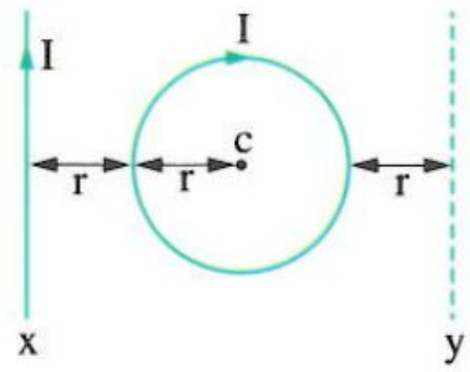
(i) The intensity of I that makes the magnetic flux density at the center of the circular coil vanish equals ..... .

(a) 35.64 A (b) 28.29 A
(c) 23.79 A (d) 20.81 A

(ii) The magnetic flux density at the center of the ring if the direction of current I is reversed equals ..... .

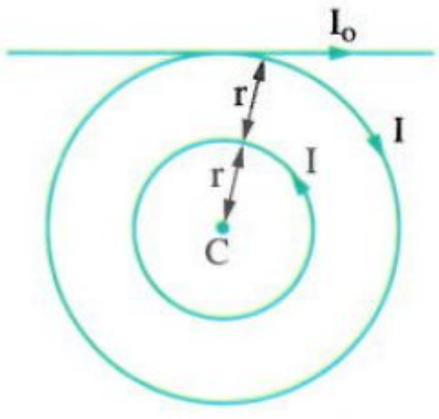
(a) 7.54×10^{-3} T (b) 3.79×10^{-3} T (c) 1.51×10^{-4} T (d) 2.24×10^{-4} T

52 ✳ The opposite figure shows a circular ring and a straight wire which is placed at position x in the same plane of the ring where both carry electric currents of intensity I, so the resultant magnetic flux density at the center of the ring c is B. If the wire is moved to position y, the magnetic flux density at point c becomes ..... .



(a) 2 B (b) B (c) 0.73 B (d) 1.38 B

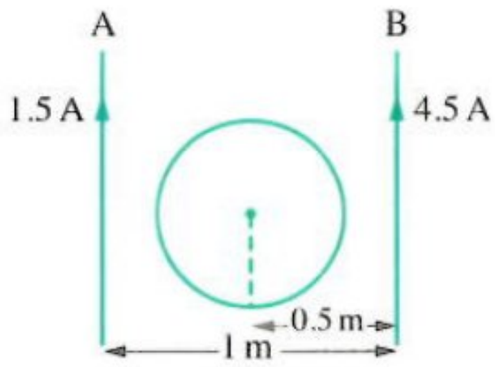
53 Two concentric rings are in the same plane of radii r, 2 r carry electric currents of intensity I in two opposite directions and a long straight wire is placed tangent to the external ring parallel to its plane and it carries an electric current of intensity I_o as shown in the opposite figure, if the resultant of the magnetic flux density at the center (C) of the two rings equals zero, then the ratio between intensities of the current $\left(\frac{I_o}{I}\right)$ equals ..... .



(a) π (b) $\frac{1}{\pi}$ (c) 2π (d) $\frac{1}{2\pi}$

Scanned with OKEN Scanner

54 ✳ Two straight parallel wires A and B are at a distance 1 m apart and carry electric currents of intensities 1.5 A and 4.5 A respectively in the same direction. A one turn coil of radius 10 π cm is placed in the same plane of the two wires where its center is at a distance of 0.5 m apart from wire A as shown in the opposite figure. So, the magnitude and the direction of the passing current in the circular coil that make the magnetic flux density at its center equals zero are ..... .



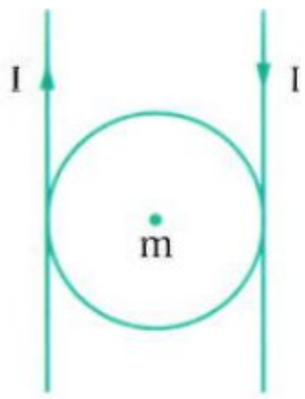
(a) 0.3 A, in the clockwise direction

(b) 0.6 A, in the clockwise direction

(c) 0.3 A, in the anti-clockwise direction

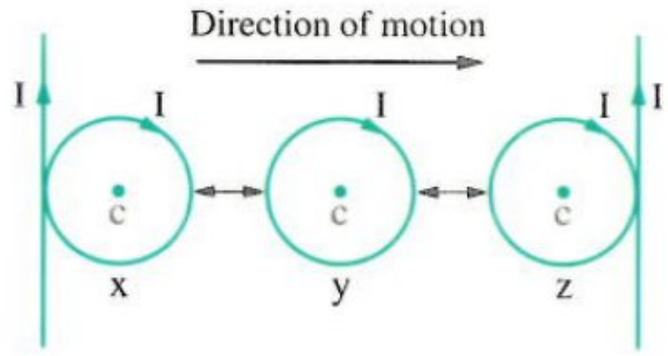
(d) 0.6 A, in the anti-clockwise direction

55 In the opposite figure, the magnetic flux density due to each one of the two straight wires at the center of the circular coil (m) is B. If the resultant of the magnetic flux density at the center of the circular coil (m) is equal to zero, then ..... .



	The direction of the current in the coil	The magnitude of the magnetic flux density due to the flow of the current in the coil
(a)	in the clockwise direction	$\frac{B}{2}$
(b)	in the anti-clockwise direction	$\frac{B}{2}$
(c)	in the clockwise direction	2 B
(d)	in the anti-clockwise direction	2 B

56 A metallic ring carries an electric current of intensity I and moves between two straight wires that carry electric currents of intensity I and are placed in the same plane of the ring as shown in the opposite figure, what is the correct order for the resultant of the magnetic flux density at the center (c) of the ring when it is fixed in each of the three positions shown in the figure?



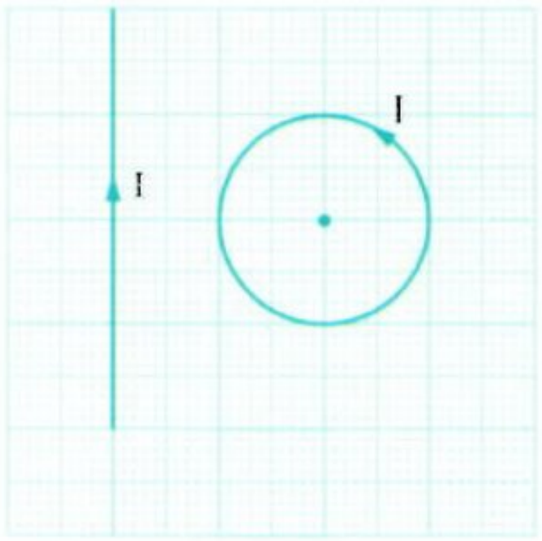
(a) $B_z > B_y > B_x$ (b) $B_x > B_y > B_z$ (c) $B_x > B_z > B_y$ (d) $B_z = B_y = B_x$

CamScanner

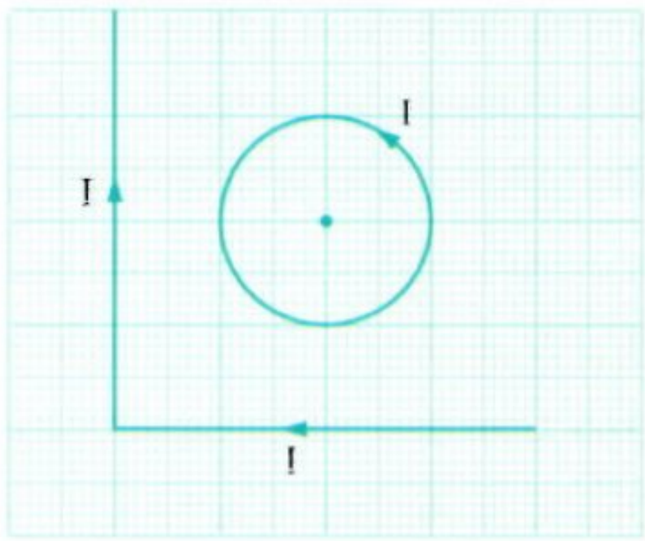
Scanned with OKEN Scanner

CamScanner

57 A metallic ring of radius r carries an electric current of intensity I and a long straight wire carrying an electric current of intensity I is placed at a distance 2r from the center of the ring in the same plane of the ring as shown in the figure (1). If the wire is bent and carries the same electric current as in figure (2):



(1)



(2)

What happens to the magnitude and the direction of the net magnetic flux density at the center of the ring?

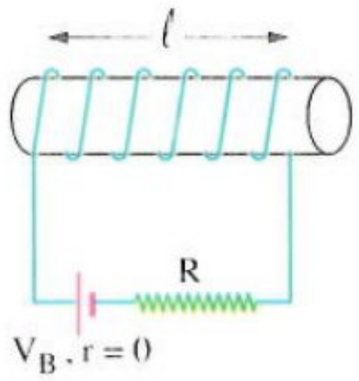
(a) Its magnitude increases and remains in the same direction.
(b) Its magnitude increases and its direction reverses.
(c) Its magnitude decreases and remains in the same direction.
(d) Its magnitude decreases and its direction reverses.

Solenoids

58 The formed magnetic field due to the flow of an electric current in a solenoid is distinguished from that of a bar magnet by the possibility of controlling ..... .

(a) the magnetic polarity only
(b) the magnetic flux density only
(c) the direction of the magnetic field only
(d) the polarity, density and direction of the magnetic field

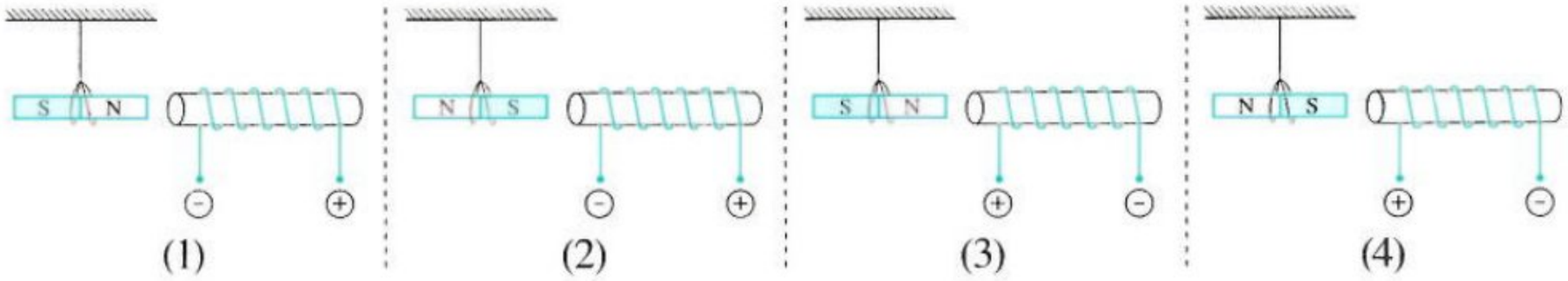
59 From the opposite figure, which of the following methods doubles the magnetic field strength inside the solenoid when the other factors are kept constant?



(a) Increasing the length (ℓ) of the coil to the double.
(b) Increasing the electromotive force (V_B) to the double.
(c) Decreasing the number of turns of the coil (N) to its half.
(d) Increasing the electric resistance of resistor R to the double.

Scanned with OKEN Scanner

60 In the following figures, a permanent bar magnet is freely hung next to a current-carrying solenoid:

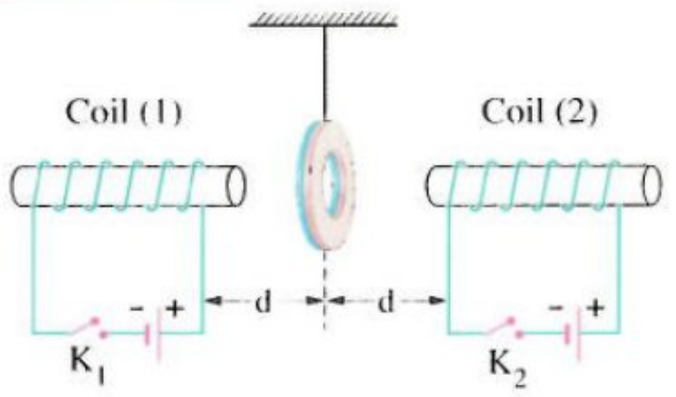


(1) (2) (3) (4)

In which two figures of them, the bar magnet gets attracted to the solenoid?

(a) (1), (2) (b) (3), (4) (c) (1), (3) (d) (2), (3)

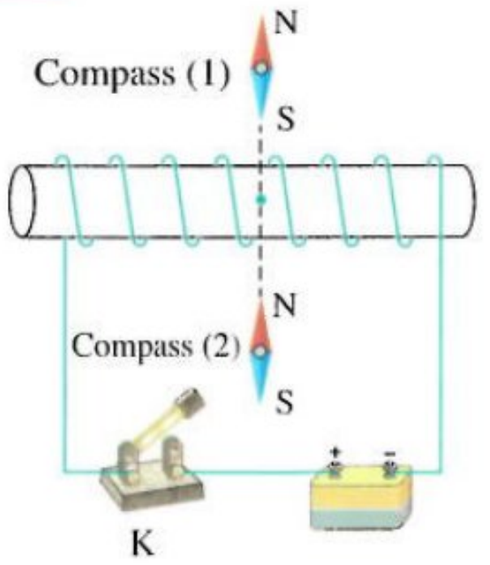
61 Two identical solenoids (1), (2), both are connected to a battery of electromotive force (V_B) of negligible internal resistance and a switch, a ring magnet was hanged at equal distances between two coils as shown in the opposite figure, after closing the two switches K_1, K_2 together, then the magnet ..... .



(a) attracts to coil (1) (b) attracts to coil (2)

(c) rotates around its axis (d) remains constant

62 Two compasses (1), (2) are placed at the two sides of a solenoid which is connected to a battery and a switch (K), if the two needles are settled by the effect of the magnetic field of the Earth as shown in the opposite figure, which of the following diagrams represents the position in which the two needles settle when the switch K is closed?



	Compass (1)	Compass (2)
(a)	N S	N S
(b)	N S	N S
(c)	N S	N S
(d)	N S	N S

63 ✳ A solenoid of length 20 cm which has 200 turns carries an electric current of 0.5 A, so the magnetic flux density at its center, if: (Take: $\mu_{iron} = 2 \times 10^{-3}$ Wb/A.m, $\pi = 3.14$)

(i) The core medium is air, equals ..... .

(a) 8.93×10^{-5} T (b) 3.11×10^{-4} T (c) 6.28×10^{-4} T (d) 7.92×10^{-5} T

(ii) An iron rod is inserted inside the coil, equals ..... .

(a) 0.5 T (b) 0.25 T (c) 0.75 T (d) 1 T

CamScanner
Scanned with OKEN Scanner

CamScanner

64 ✱ A solenoid of length 0.22 m and cross-sectional area 25×10^{-4} m^2 consists of 300 turns. An electric current passes through the solenoid so that a magnetic flux density at its center becomes 1.2×10^{-3} Wb/m^2, hence:

(i) The electric current intensity equals ..... .

(a) 0.3 A (b) 0.5 A (c) 0.7 A (d) 1.1 A

(ii) The total magnetic flux through a cross-section of the solenoid equals ..... .

(a) 3×10^{-6} Wb (b) 5×10^{-6} Wb (c) 9×10^{-6} Wb (d) 18×10^{-6} Wb

---

65 ✱ A solenoid of length 0.6 m carries an electric current of 10 A, hence the magnetic flux density that is produced at its center equals 0.05 T, so:

(i) The number of turns per unit length of the solenoid equals ..... .

(a) 2388.5 turns/m (b) 3679.4 turns/m (c) 3977.3 turns/m (d) 5123.5 turns/m

(ii) The number of turns of the solenoid equals ..... .

(a) 1194.5 turns (b) 2386.4 turns (c) 3582.7 turns (d) 8359.3 turns

---

66 A solenoid of N turns has a length of 50 cm and the resistance of each one turn is 0.01 Ω. The solenoid is connected to an electric source whose emf is 2 V that has a negligible internal resistance, so:

(i) The magnetic flux density at its center equals ..... .

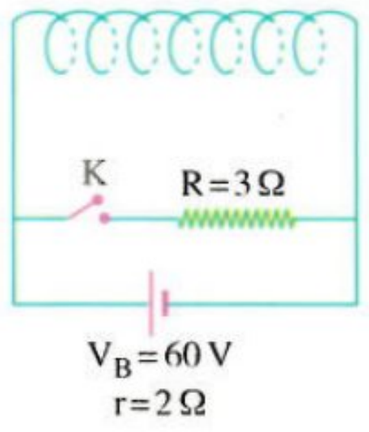
(a) 2.15×10^{-3} T (b) 4.11×10^{-3} T (c) 3.17×10^{-4} T (d) 5.03×10^{-4} T

(ii) The magnetic flux density, if half of the number of the solenoid turns has been cut off and its other half is connected to the same source, becomes ..... .

(a) 1.01×10^{-3} T (b) 3.09×10^{-3} T (c) 2.17×10^{-4} T (d) 7.18×10^{-4} T

---

67 ✱ A solenoid of length 20 cm, number of turns 100 turns and resistance 6 Ω is inserted in the electric circuit as shown in the opposite figure, hence the magnetic flux density at its center in the case of:



(i) opening switch K equals ..... .

(a) 3.14×10^{-3} T (b) 4.71×10^{-3} T

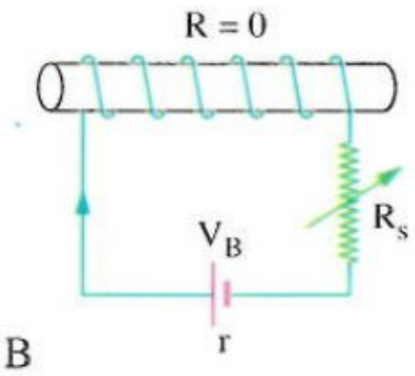
(c) 6.22×10^{-3} T (d) 9.78×10^{-3} T

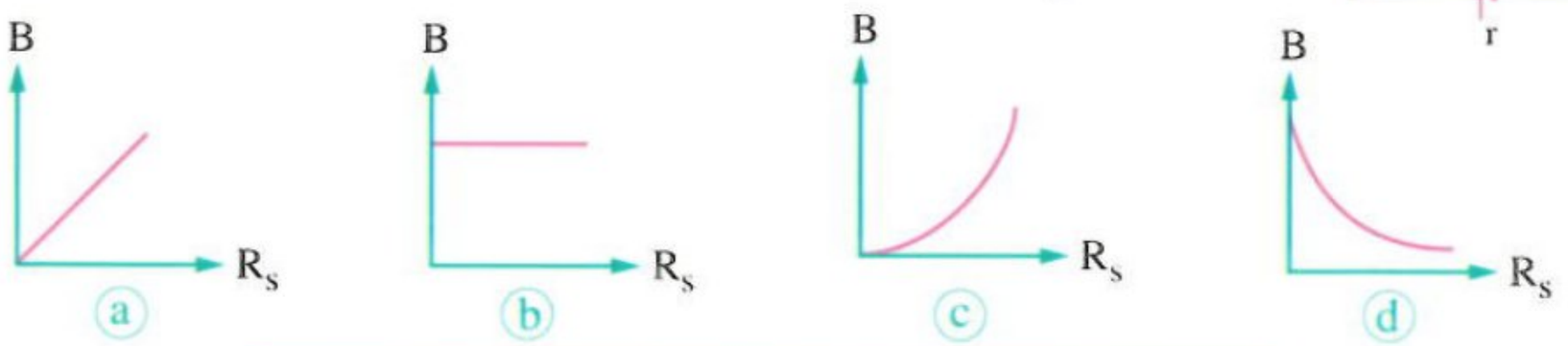
(ii) closing switch K equals ..... .

(a) 3.14×10^{-3} T (b) 5.19×10^{-3} T (c) 6.03×10^{-3} T (d) 7.16×10^{-3} T

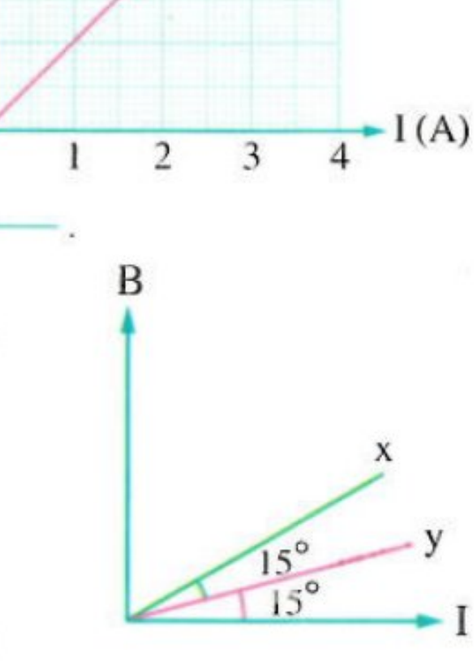
Scanned with OKEN Scanner

68 The opposite figure shows a solenoid of negligible resistance connected in an electric circuit, so which of the following graphs represents the variation of the magnetic field (B) at the center of the solenoid versus the value of the variable resistance (R_s)?





69 ✱ In the opposite electric circuit, what happens to the magnitude of the magnetic flux density inside the solenoid due to an electric current passing through the coil when increasing the value of:



(i) resistance R_1?

(a) Increases. (b) Decreases.

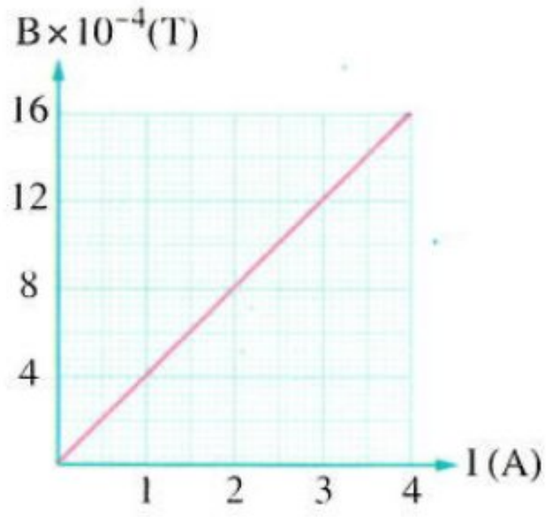
(c) Remains unchanged. (d) Indeterminable.

(ii) resistance R_2?

(a) Increases. (b) Decreases.

(c) Remains unchanged. (d) Indeterminable.

70 The opposite graph represents the relation between the magnetic flux density (B) at the center of a solenoid and the current intensity (I) that flows in the solenoid. So, the number of turns per meter for this solenoid equals ..... .



(a) 215.2 turns/meter

(b) 250.5 turns/meter

(c) 318.18 turns/meter

(d) 341.4 turns/meter

71 Two solenoids x and y have the same number of turns, but their cores are made of two different materials. Solenoid y is four times longer than the solenoid x, and both carry electric currents. The opposite graph represents the relation between the magnetic flux density (B) at the center of each solenoid along its axis and the current intensity (I) passing through it. Which of the two solenoid cores has the material with the greater magnetic permeability?

(a) The core of solenoid x (b) The core of solenoid y

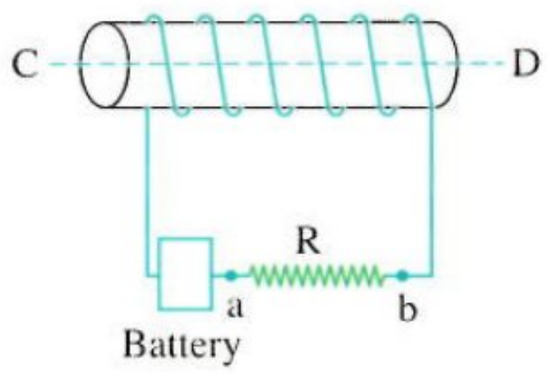
(c) The magnetic permeability of the materials used in the cores of the two solenoids are equal.

(d) The answer can't be determined.

CamScanner
Scanned with OKEN Scanner

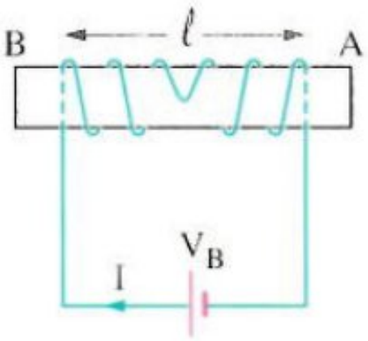
CamScanner

72 In the opposite figure, a solenoid of length 10 π cm whose number of turns is 500 turns, is connected to a battery and a resistor R in series, hence a south pole is formed at end D of the solenoid where the magnetic flux density at its center is 2.4×10^{-2} T, so the intensity and the direction of the current in the resistor R are ..... .



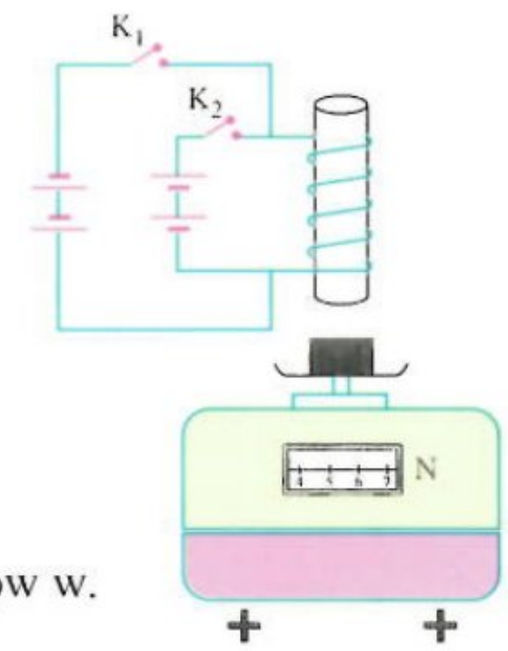
(a) 12 A, from a to b
(b) 12 A, from b to a
(c) 24 A, from a to b
(d) 24 A, from b to a

73 In the opposite figure, a wire is wound to form a long solenoid and then it is connected to a battery, so the formed poles at terminals A, B respectively are ..... .



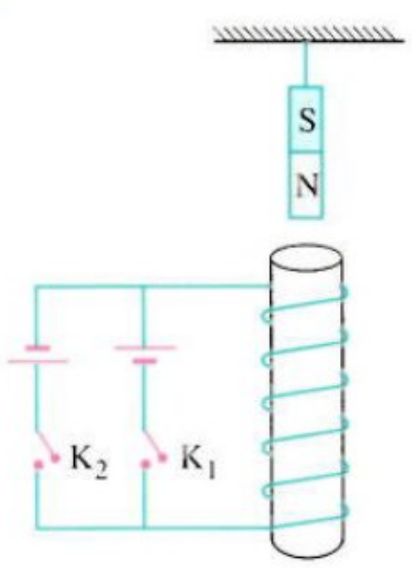
(a) north, north
(b) north, south
(c) south, north
(d) south, south

74 In the opposite figure, a coil is fixed above a piece of soft iron that is placed on the pan of a scale which gives a reading value w, what happens to the reading of the scale when one of the two switches K_1 or K_2 is closed while keeping the other one open?



(a) Increases in both cases above w.
(b) Decreases in both cases below w.
(c) Increases in one case above w and decreases in the other below w.
(d) Remains w in both cases.

75 In the opposite diagram, a solenoid of resistance R is connected to two identical batteries through two switches K_1, K_2 . A magnet is freely hanged by a thread above the solenoid. Which of the following actions would reduce the tension in the thread?



(a) Closing switch K_1 only.
(b) Closing switch K_2 only.
(c) Closing the two switches K_1, K_2 together.
(d) The answer can't be determined.

Scanned with OKEN Scanner

76 ✳ An insulated wire of diameter 0.2 cm is coiled around an iron rod of permeability 2×10^{-3} Wb/A.m such that the turns of wire are tangent to each other along the rod, so if an electric current of 1 A passes through the turns of wire, the magnetic flux density at its center equals ..... .

(a) 0.5 T (b) 1 T (c) 1.2 T (d) 1.5 T

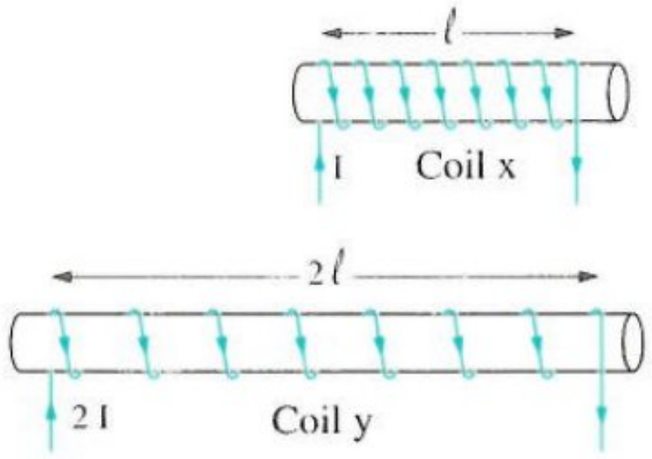
77 ✳ A uniformly coiled solenoid of length ℓ and number of turns N is cut into two parts x and y of lengths ℓ_1 and $3\ell_1$ respectively and each of them is connected to the same electric potential difference, so the ratio between the magnetic flux densities $\left(\frac{B_x}{B_y}\right)$ at the centers of the two coils equals ..... .

(a) $\frac{1}{1}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{3}{1}$

78 Two solenoids are identical, the first one is made of copper wire and the second is made of aluminum wire and each of them was connected separately one at a time to the same battery, so the magnetic flux densities at the center of each of them due to the flow of an electric current in them were B_1 and B_2 respectively, then .....
(Where the resistivity of copper is less than the resistivity of aluminum)

(a) $B_1 > B_2$ (b) $B_1 < B_2$ (c) $B_1 = B_2 = 0$ (d) $B_1 = B_2 \neq 0$

79 If two solenoids (x and y) that have the same number of turns and are wound around iron cylinders, carry direct electric currents as shown in the opposite figure, then the ratio of the produced magnetic flux densities at the axis inside each of them $\left(\frac{B_x}{B_y}\right)$ equals ..... .



(a) $\frac{2}{1}$ (b) $\frac{1}{4}$
(c) $\frac{1}{1}$ (d) $\frac{1}{2}$

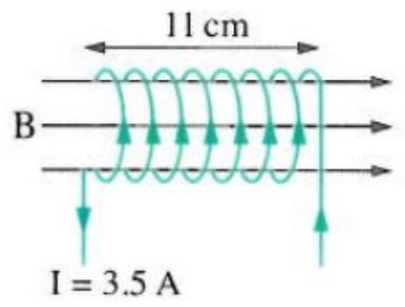
80 ✳ Two solenoids X,Y of the same length and the same number of turns are made of two copper wires of the same length but of different cross-sectional areas and they are connected to two sources of the same voltage, if the ratio of the magnetic flux densities at the center of each of them is $\frac{B_X}{B_Y} = \frac{9}{1}$, which of the following choices is correct?

(a) The cross-sectional area of wire X is three times that of wire Y
(b) The cross-sectional area of wire X is nine times that of wire Y
(c) The resistance of wire X is three times that of wire Y
(d) The resistance of wire X is double that of wire Y

CamScanner
Scanned with OKEN Scanner

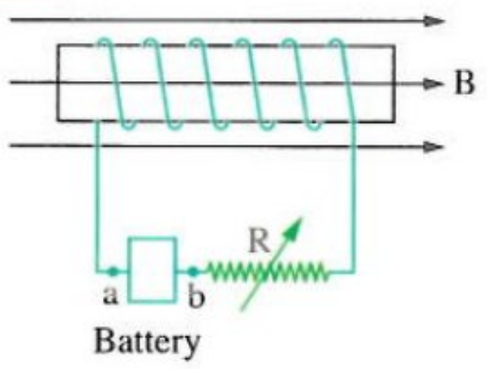
CamScanner

81 ✱ The opposite figure shows a solenoid of 60 turns that carries an electric current. If this coil is placed completely in an external magnetic field of flux density 5.2×10^{-3} T whose direction is parallel to the axis of the solenoid and to the right of the page, then at the center of the solenoid, the net magnetic flux density and its direction are ..... .



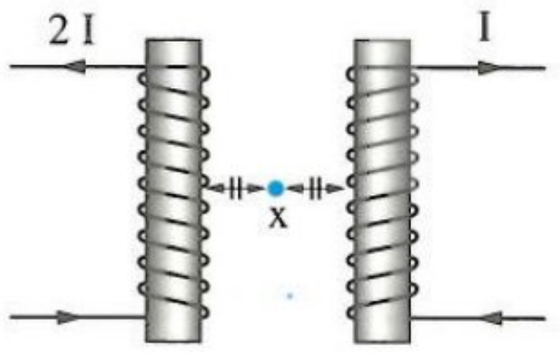
(a) 2.8×10^{-3} T, in the direction of the external field
(b) 2.8×10^{-3} T, opposite to the direction of the external field
(c) 7.6×10^{-3} T, opposite to the direction of the external field
(d) 7.6×10^{-3} T, in the direction of the external field

82 ✱ In the opposite figure, a solenoid of length 1.1 m whose number of turns is 210 turns is placed in an external magnetic field of flux density 1.2×10^{-3} T that is parallel to the solenoid axis, so the electric current that should flow through the circuit and the poles of the battery, that make the resultant magnetic flux density at the center of the solenoid become zero, are ..... .



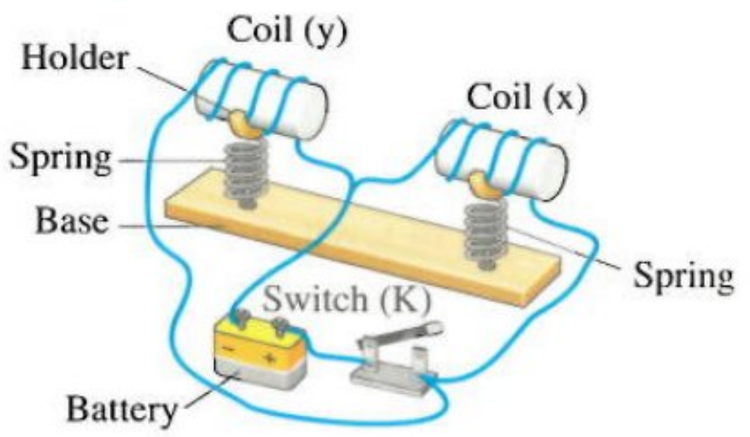
	The electric current that passes through the circuit	The poles of the battery
(a)	8 A	a is +ve pole, b is –ve pole
(b)	8 A	a is –ve pole, b is +ve pole
(c)	5 A	a is +ve pole, b is –ve pole
(d)	5 A	a is –ve pole, b is +ve pole

83 In the opposite figure, two identical solenoids, each of them carries an electric current, are placed near to each other where their axes are parallel, so the net magnetic field due to them at point x is directed ..... .



(a) into the page
(b) out of the page
(c) upwards
(d) downwards

84 A student designed a device to study the magnetic field produced due to the flow of an electric current through two identical solenoids x, y as in the opposite figure, what do you expect to happen when the switch (K) is closed?



(a) Repulsion between two coils.
(b) Attraction between two coils.
(c) No magnetic force is generated between two coils.
(d) The two coils vibrate vertically.

Scanned with OKEN Scanner

85 ✳ Two solenoids are put one inside the other where their axes coincide with each other. Each unit of length of the first solenoid contains 10 turns and that of the second solenoid contains 20 turns, so if the passing electric current in the first one is 2 A while that in the second one is 4 A, the magnetic flux density at a point on the axis inside them:

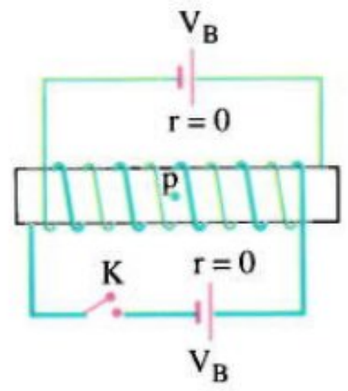
(i) when the two electric currents are in the same direction equals ..... .

(a) 125.71×10^{-6} T (b) 432.7×10^{-5} T (c) 79.7×10^{-4} T (d) 45.31×10^{-3} T

(ii) when the two electric currents are in opposite directions equals ..... .

(a) 96.41×10^{-6} T (b) 85.7×10^{-6} T (c) 75.43×10^{-6} T (d) 41.8×10^{-6} T

86 The opposite figure shows two identical solenoids, so after closing switch K, the net magnetic flux density at their common center (point p) ..... .



(a) increases (b) decreases

(c) doesn't change (d) becomes zero

87 A circular coil of N turns and radius r carries electric current I, hence the magnetic flux density at its center is B. If its turns are displaced uniformly away from each other to become a solenoid of length 20 r while carrying the same current, so the magnetic flux density at its center becomes ..... .

(a) $\frac{B}{20}$ (b) $\frac{B}{10}$ (c) $\frac{B}{40}$ (d) B

88 A circular coil of radius 2.2 cm carries an electric current, hence a magnetic flux of density B is produced at its center, so if its turns are drawn away uniformly from each other till it becomes 110 cm long, the magnetic flux density at its center will equal ..... .

(a) B (b) 2 B (c) 0.5 B (d) 0.04 B

89 ✳ A solenoid of diameter 10 cm and length 200 cm carries an electric current that produces a magnetic flux of density 4×10^{-6} T at its center, so if its turns are compressed uniformly till it becomes a circular coil, the magnetic flux density at the coil's center in this case becomes ..... .

(a) 2.75×10^{-3} T (b) 4.5×10^{-3} T (c) 8×10^{-5} T (d) 13×10^{-3} T

90 ✳ A solenoid of length 50 cm and number of turns 100 turns carries an electric current of 2 A. A circular coil is placed exactly at the middle of the solenoid such that the circular coil and the solenoid become coaxial. If the number of turns of the circular coil is 20 turns while its radius is 15 cm and it carries electric current of 1 A, the magnetic flux density at the center of the circular coil, if the two currents are:

CamScanner

Scanned with OKEN Scanner

CamScanner

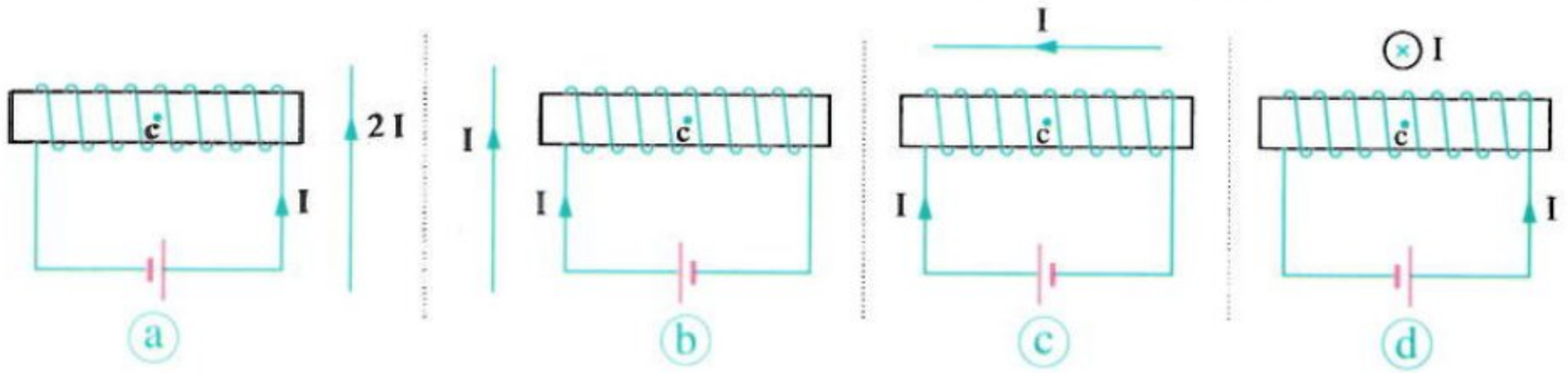
(i) In the same direction, equals ..... .

(a) 3.32×10^{-4} T (b) 4.19×10^{-4} T (c) 5.87×10^{-4} T (d) 6.93×10^{-4} T

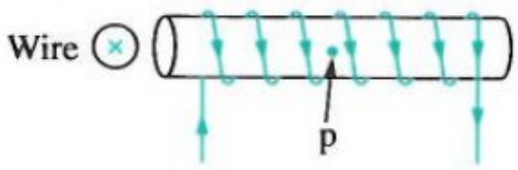
(ii) In opposite directions, equals ..... .

(a) 4.19×10^{-4} T (b) 9.73×10^{-3} T (c) 8.62×10^{-4} T (d) 12.5×10^{-4} T

91 **In which of the following figures, the net magnetic flux density at the center of the solenoid (point c) is leftwards and parallel to the plane of the page?**

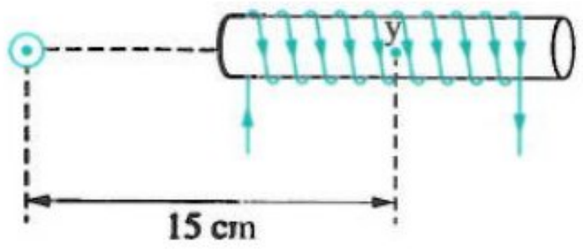


92 ✳ **In the opposite figure, a solenoid carries an electric current that generates a magnetic flux density B at its center (point p) and beside the solenoid there is a straight wire that is perpendicular to the plane of the page where it carries a current that generates a magnetic flux density B at point p. So, the total magnetic flux density at point p is** ..... .



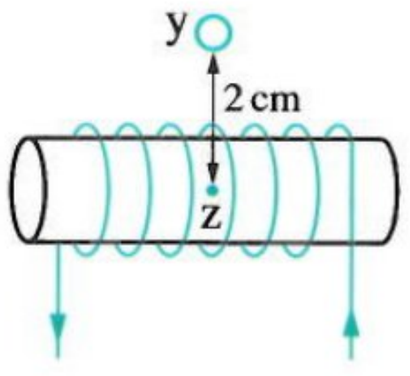
(a) zero (b) B (c) $\sqrt{2}$ B (d) 2 B

93 ✳ **A straight wire carrying electric current of 15 A is placed perpendicular to the axis of a solenoid of 10 turns whose length is 15 cm which carries an electric current of $\frac{7}{22}$ A as shown in the opposite figure, so the magnetic flux density at point y which is at a distance 15 cm from the mid-point of the wire's axis equals** ..... .



(a) 11.5×10^{-3} T (b) 3.34×10^{-5} T (c) 5.67×10^{-4} T (d) 9.7×10^{-6} T

94 ✳ **A straight wire (y) is perpendicular to the plane of the page at a distance 2 cm from the axis of a solenoid that consists of 50 turns/m and carries a current of intensity 1.4 A as in the opposite figure, so the net magnetic flux density at the center of the solenoid (point z) vanishes, when the direction and the intensity of the current that passes in the wire are** ..... .

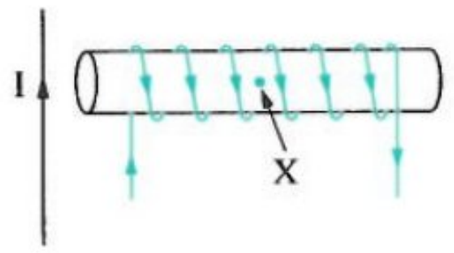


(a) perpendicular into the page, 2.2 A (b) perpendicular into the page, 6.6 A

(c) perpendicular out of the page, 8.8 A (d) perpendicular out of the page, 4.4 A

Scanned with OKEN Scanner

95 ✱ In the opposite figure, when an electric current passes through the solenoid, it generates a magnetic flux density of 3×10^{-6} T at the center of the solenoid (at point X). If a straight wire is placed close to one of the solenoid ends and in the same plane of its axis where an electric current passes through it and generates a magnetic flux density of 4×10^{-6} T at point X, so the resultant magnetic flux density at point X is ..... .

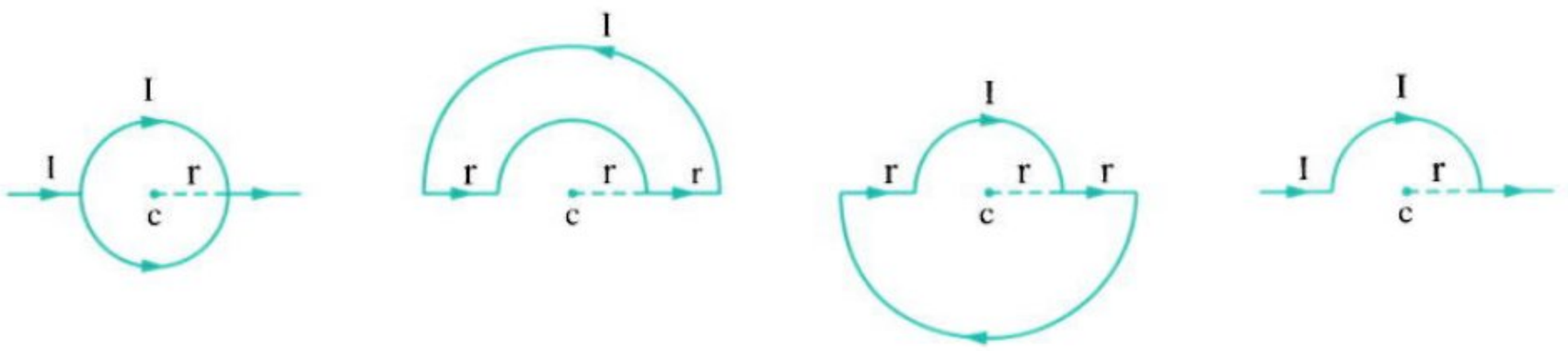


(a) 10^{-6} T (b) 5×10^{-6} T (c) 7×10^{-6} T (d) 12×10^{-6} T

Second Essay questions

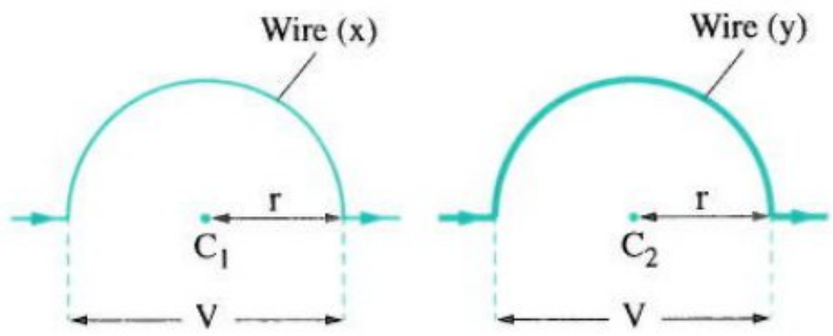
Circular coils

1 Many wires are shaped as the following figures and electric current passes in each of them.
Arrange in ascending order the magnetic flux density B at center (c) in each case.



(1) (2) (3) (4)

2 Two wires (x, y) are made of the same material and having the same length, the ratio between their cross-sectional areas respectively is $\frac{1}{2}$. The two wires are shaped so each one becomes a semi-circular ring (1), (2) as in the opposite figure. If the potential difference across the ends of each of them is V, **then**



(1) (2)

calculate the ratio between the magnetic flux densities $\left(\frac{B_x}{B_y}\right)$ at the centers C_1, C_2

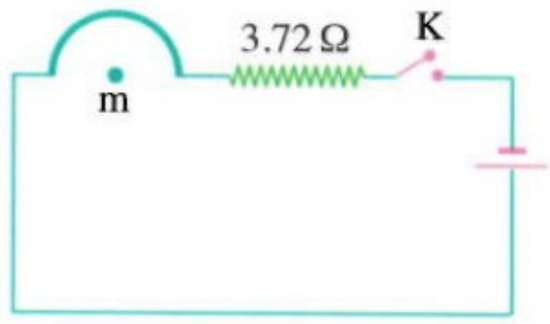
3 An electric current is set up in a circular coil of number of turns N_1, so a magnetic flux of density B_1 is formed at its center. When the coil is recoiled to be a circular coil of number of turns N_2 in which the same current intensity is set up, the magnetic flux density at its center becomes B_2. **Prove that:** $\frac{B_1}{B_2} = \left(\frac{N_1}{N_2}\right)^2$

CamScanner
Scanned with OKEN Scanner

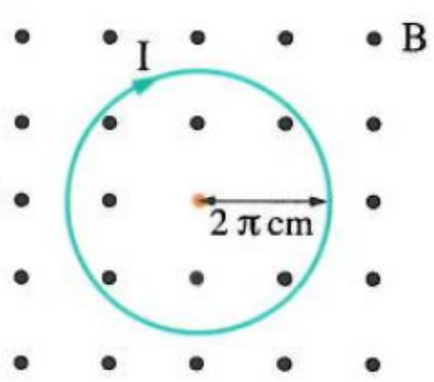
CamScanner

4 A circular coil of radius 3.14 cm where the resistance of one turn of it is 2 Ω, its terminals are connected to a DC source of potential difference 4 V. **Calculate** the magnetic flux density at the center of the coil.

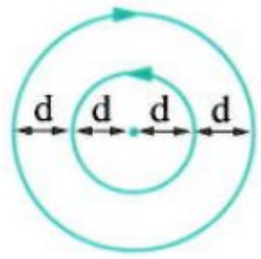
5 In the opposite figure, a wire of resistance R in the form of half a circular ring of radius 3.14 cm, connected in series with a resistor of 3.72 Ω and a source of emf 24 V with internal resistance of 2 Ω where the connecting wires are of negligible resistance. When the switch K was closed, the magnetic flux density at the center of the half circular ring (m) due to the flow of current through it was 2.4×10^{-5} T, **calculate** the resistance of the wire (R). (Where: $\pi = 3.14$)



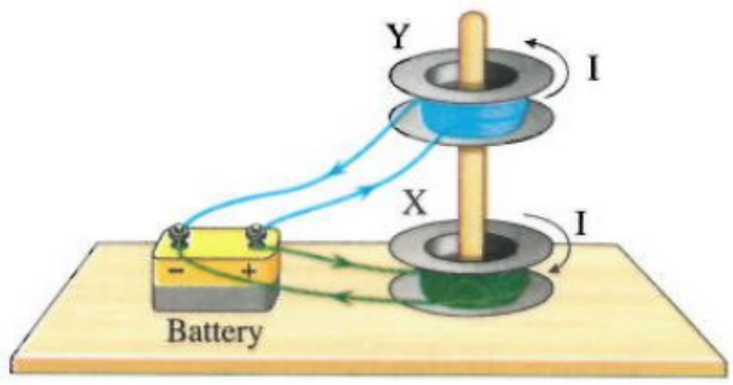
6 An external magnetic field (B) of flux density 6×10^{-5} T acts perpendicularly on the plane of a metal ring of radius 2 π cm that carries electric current of intensity I, in the direction which is shown in the opposite figure. If the net magnetic flux density at the center of the ring is 4×10^{-5} T and its direction is in the direction of the external magnetic field B. **Calculate** the current intensity I.



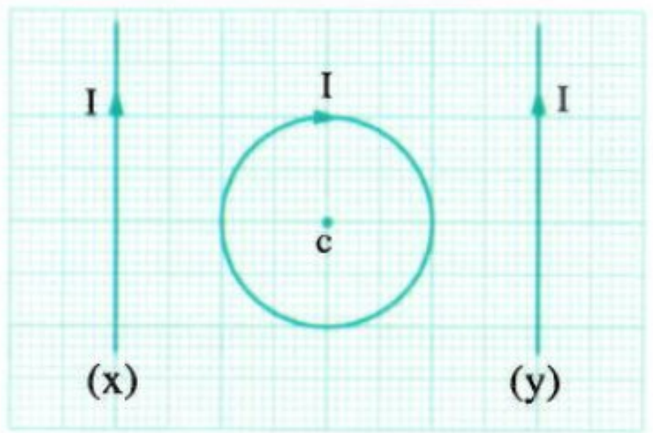
7 Two concentric copper rings in the same plane carry the same electric current intensity (I) as in the opposite figure. **What** change is required in the current intensity of the inner ring to make the common center point a neutral point?
Explain your answer.



8 Two circular coils X, Y are identical, coil Y is placed free to move above coil X and each of them carries electric current of intensity I whose directions when viewed from the upper face of the two coils are as shown in the opposite figure. **Explain why** coil Y didn't fall and remained suspended.



9 A metallic ring carries an electric current I, where two straight wires are placed in its plane and carry the same current I as in the opposite figure. **Explain what happens** to the magnitude of the net magnetic flux density at the center of the ring (c), if the electric current intensity has increased to double in:
(a) wire x only. (b) wire y only.



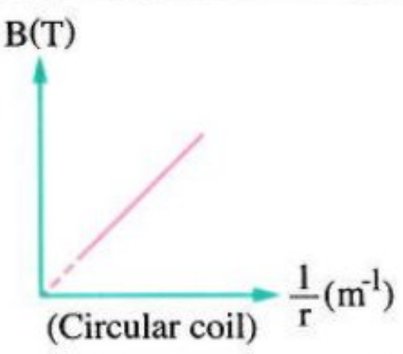
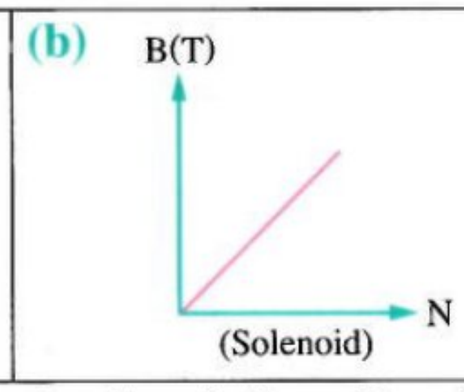
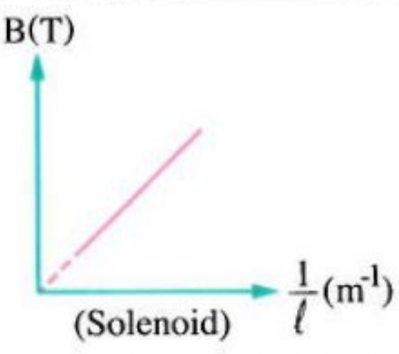
Scanned with OKEN Scanner

Solenoids

10 The resistance of one turn of a solenoid of length ℓ is R. The ends of the solenoid are connected to a DC electric source of potential difference V. **Prove that** the magnetic flux density at the center of the solenoid is calculated from the relation; $B = \frac{\mu V}{\ell R}$

11 A solenoid with an air core carries a DC electric current, **mention four suggestions** each of them individually leads to increasing the magnetic flux density at the center of a solenoid.

12 **Write the mathematical relation and the equivalent of the slope of the line for each of the following graphs:**

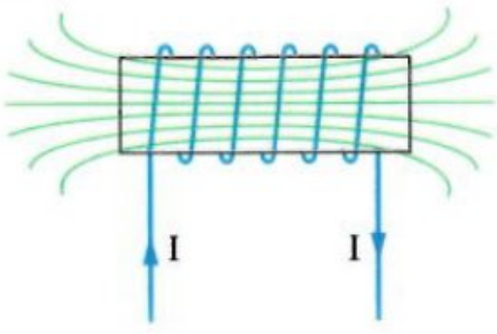
(a)	(b)	(c)
		

(Where: B is the magnetic flux density, r is the radius of the coil, N is the number of turns and ℓ is the length of the solenoid)

13 A solenoid of resistance R is tightly coiled such that its turns are touching each other, its terminals are connected to a battery of negligible internal resistance. **How does** the net magnetic flux density at the center of the solenoid get affected after doing the following changes:

- **(a)** Moving the solenoid turns uniformly away from each other.
- **(b)** Reshaping the turns of the coil by increasing their diameter to be double while keeping them touching each other.
- **(c)** Connecting a similar coil in series along with the axis of the first coil to become one coil while keeping all of the turns touching each other, then connecting its free ends to the same battery.

14 The opposite figure illustrates the magnetic field formed due to an electric current in a solenoid:



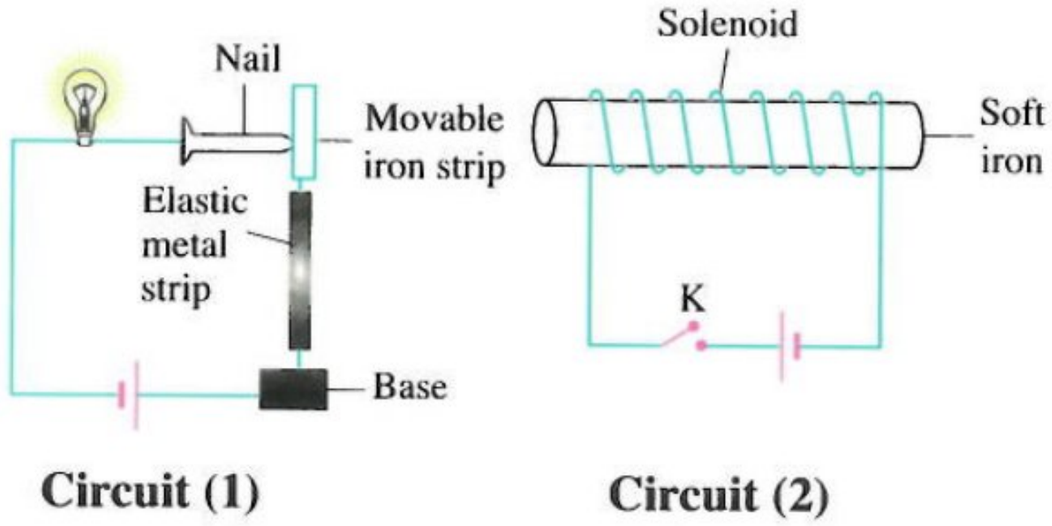
- **(a)** **Determine with arrows, on the figure** the direction of the magnetic flux lines.
- **(b)** **Why** is the magnetic flux density outside the ends of the coil less than that at its center?

CamScanner

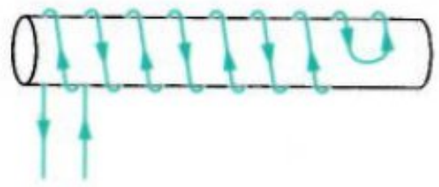
Scanned with OKEN Scanner

CamScanner

15 In the following figure, **explain why** the electric bulb turns off in circuit (1) when the switch K is closed in circuit (2).

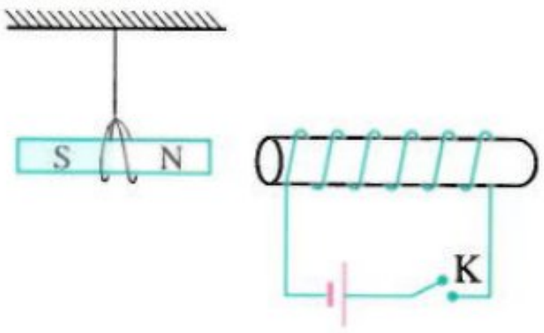


16 In the opposite figure, a solenoid carries an electric current and coiled on an iron bar. **Explain why** the iron bar is not magnetized.



17 In the opposite figure:

A solenoid is coiled around a plastic cylinder and connected to an electric source where a freely swinging magnet is suspended near them as shown in the figure:

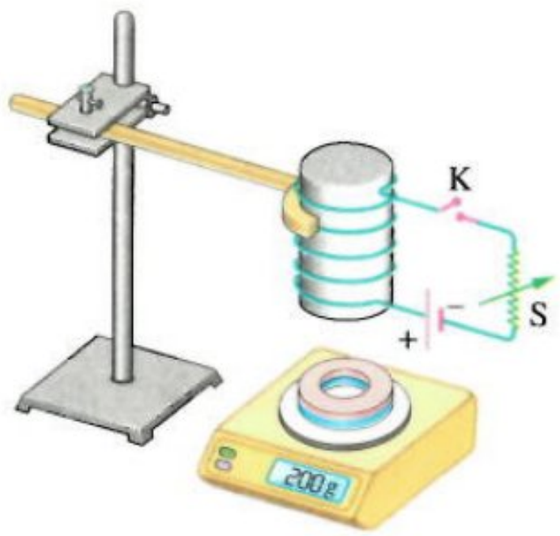


(a) **What** is the type of the magnetic force which is acting on the N pole of the magnet when the switch (K) is closed?

(b) **What will happen to** the magnitude of the magnetic force when the plastic cylinder is replaced by a cylinder made of soft iron then the switch is closed?

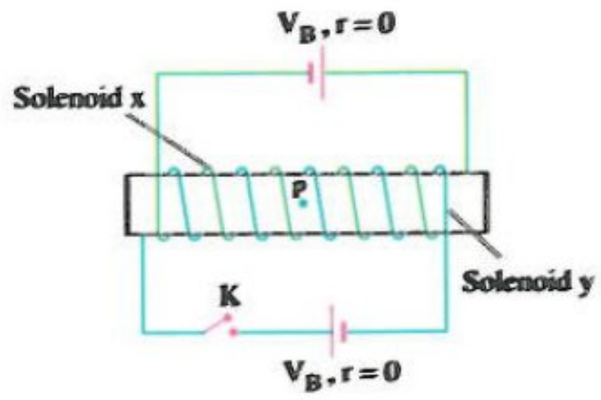
(c) **What will happen to** the type of the magnetic force when the terminals of the electric source are reversed then the switch is closed?

18 In the opposite figure, a solenoid is connected to a switch (K), a variable resistor (S) and an electric cell. The coil is fixed above a balance pan on which a short magnet is placed such that its upper face is a north pole. **Explain what happens to** the reading of the balance after closing the switch (K).



Scanned with OKEN Scanner

19 The opposite electric circuit illustrates two identical solenoids x, y but solenoid x is made of aluminum, while y is made of copper. Since the resistivity of aluminum is greater than that of copper, **what happens to** the net magnetic flux density at center of the two solenoids (point p) when closing the switch K which is connected to solenoid y?



20 A circular coil of diameter 12 cm, carries an electric current that produces a magnetic field at its center. The turns of the coil are displaced uniformly from each other to become a solenoid that carries the same electric current. So, the magnetic flux density at its center has become $\frac{1}{4}$ that at the center of the circular coil. **Calculate** the length of the solenoid.

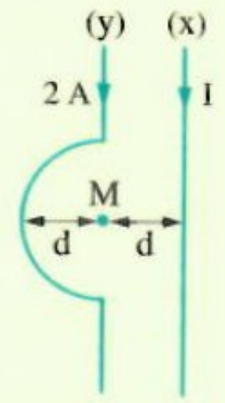
CamScanner
Scanned with OKEN Scanner

CamScanner

Questions of the Previous Years Exams on Lesson 2

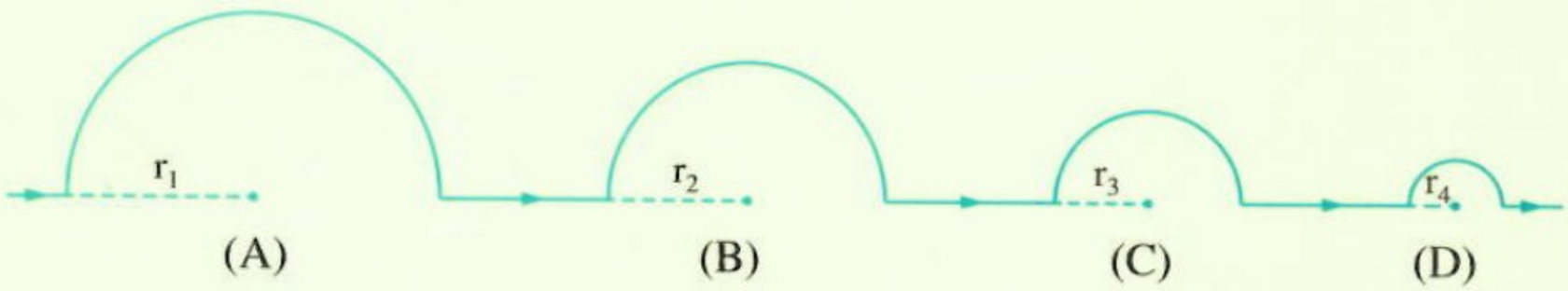
Circular coil

1 The figure illustrates two current-carrying conductors (x) and (y). A current of intensity I passes through conductor (x) while a current of intensity (2 A) passes through conductor (y). Then, the current (I) that causes the magnetic flux density at point M to be zero is ..... . *(Exp. May 21)*



(a) $\frac{\pi}{2}$ A (b) $\frac{\pi}{4}$ A (c) 2π A (d) π A

2 The figure illustrates a wire in the form of semi-circles connected together and connected to a cell. *(Exp. May, June 21)*



Which of these semi-circles has the least magnetic flux density at its center?

(a) A (b) B (c) C (d) D

3 A straight wire was wound in the form of a circular coil of N turns and a current of intensity I was set up in it. If it is reformed so that the number of turns becomes $\frac{N}{4}$ and the same current intensity is set up in it, the magnetic flux density at its center will be ..... of its initial value. *(Exp. June 21)*

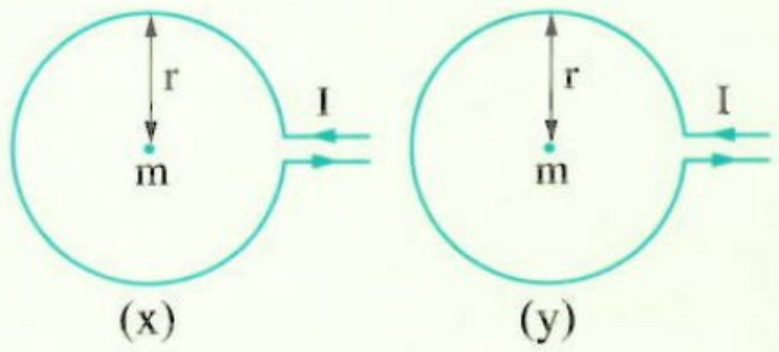
(a) $\frac{1}{16}$ (b) 16 times (c) 4 times (d) $\frac{1}{4}$

4 A straight wire was wound to form a circular coil of N turns, through which a current of intensity I was flowing to form a magnetic flux of density B at the center of the coil. If the same wire was rewound so that the number of turns became $\frac{2N}{3}$ and the same current intensity I passes through it, the magnetic flux density at its center would become ..... . *(1st session 21)*

(a) $\frac{2}{3}$ B (b) $\frac{2}{9}$ B (c) $\frac{1}{9}$ B (d) $\frac{4}{9}$ B

Scanned with OKEN Scanner

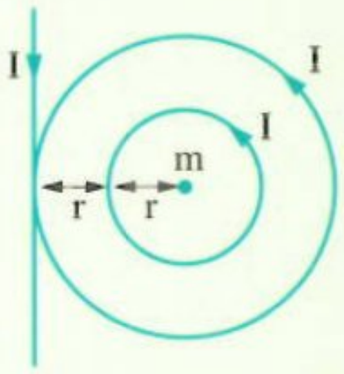
5 In the opposite figure, two circular coils (x) and (y) having the same diameter while electric currents of the same intensity (I) pass through each of them. If the number of turns of coil (x) is double the number of turns of coil (y), which of the following relations represents correctly the relation between the produced magnetic flux densities (B) at the center of each coil?



(1st session 21)

(a) $B_{(x)} = 2\,B_{(y)}$ (b) $B_{(x)} = B_{(y)}$ (c) $B_{(x)} = \frac{1}{2}\,B_{(y)}$ (d) $B_{(x)} = 4\,B_{(y)}$

6 A current-carrying straight wire is put tangent to the outer ring of two concentric rings that carry electric currents in the same direction. If all of them are in the same plane and carry currents that have intensity I as shown in the figure, the resulting magnetic flux density at the common center (m) due to the three currents equals .....



(1st session 21)

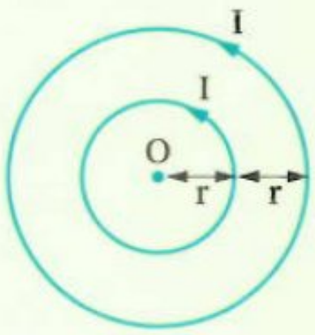
(a) $\frac{0.83\,\mu I}{r}$ (b) $\frac{0.67\,\mu I}{r}$ (c) $\frac{0.54\,\mu I}{r}$ (d) $\frac{0.42\,\mu I}{r}$

7 A circular coil of number of turns N and radius r carries a current of intensity I, so a magnetic field of flux density B_1 is generated at its center. If the coil is connected to another electric source such that a current of intensity which equals three times its intensity in the first case passes through it, a magnetic field of flux density B_2 is generated at its center. So, ...

(2nd session 21)

(a) $B_2 = 3\,B_1$ (b) $B_2 = B_1$ (c) $B_2 = \frac{1}{3}\,B_1$ (d) $B_2 = \frac{3}{2}\,B_1$

8 Two concentric rings both carry electric current I in the same direction as shown in the figure, producing magnetic field of flux density B at the common center (O). If the direction of the passing current through one ring only is reversed keeping the other current as it is, so the resultant magnetic flux density at (O) becomes ..... .



(2nd session 21)

(a) $\frac{B}{2}$ (b) $\frac{B}{4}$ (c) $\frac{B}{3}$ (d) $\frac{B}{5}$

9 A circular coil of number of turns (N) and radius (r) is carrying an electric current of intensity (I), so a magnetic field of flux density (B) is generated at its center. If the coil is cut such that a quarter of its turns are removed and the same current (I) is set to pass through it. So, the magnetic flux density at the center of the coil in the second case equals ..... .

(1st session 22)

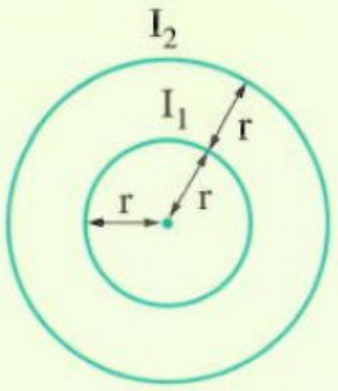
(a) B (b) $\frac{3}{4}$ B (c) $\frac{3}{2}$ B (d) $\frac{4}{3}$ B

CamScanner

Scanned with OKEN Scanner

CamScanner

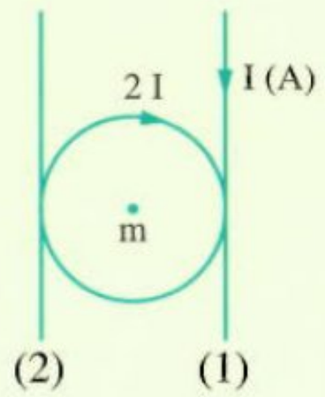
10 The figure illustrates two concentric circular coils of same number of turns and having different radii, each coil produces magnetic flux of density (B) at the common center due to the electric current passing through each of them (I_1, I_2) as shown. Which of the following choices correctly represents the relation between the values of (I_1 and I_2), their directions and also the magnitude of the resultant magnetic flux density (B_t) at the common center?



(2nd session 22)

	The relation between the values of I_1, I_2 and their directions	B_t
(a)	$I_1 = I_2$, in the same direction	2 B
(b)	$I_2 = 2\ I_1$, in the opposite direction	zero
(c)	$I_2 = I_1$, in the opposite direction	zero
(d)	$I_2 = \frac{1}{2}\ I_1$, in the same direction	2 B

11 A metallic ring was carrying electric current of intensity (2 I), a magnetic flux of density (B) was generated at the center of the ring (m). If two straight current-carrying wires (1) and (2) were placed tangent to the ring and in the same plane as in figure.
So, to make the resultant magnetic flux density due to the three electric currents have the same value (B) at the center (m), the electric current intensity that passes in the wire (2) = ... and its direction ..... .

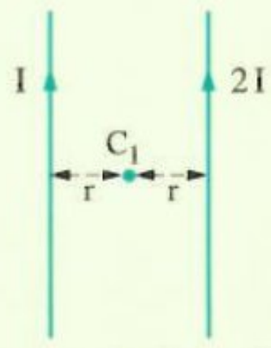


(2nd session 22)

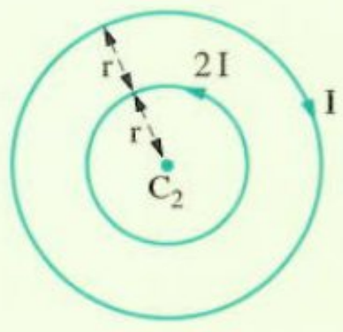
(a) (I), to the top of the page
(b) (I), to the bottom of the page
(c) (2 I), to the bottom of the page
(d) (2 I), to the top of the page

12 Using the data illustrated in both figures (1) and (2). Which of the following relations represents correctly the relation between the net magnetic flux densities (B) at points C_1 and C_2?

(1st session 23)



Two long parallel straight wires
(1)

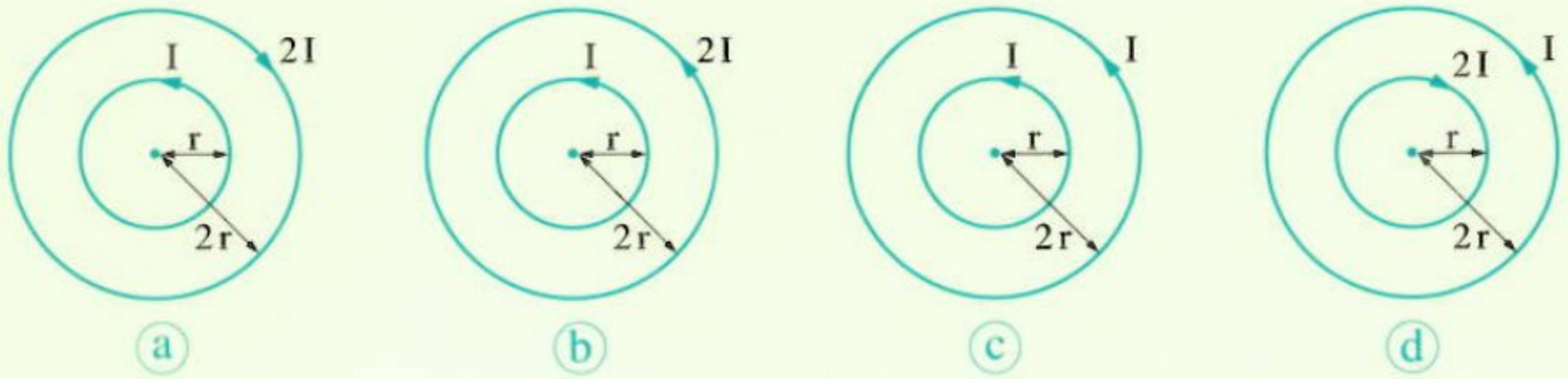


Two metallic coaxial rings
(2)

(a) $B_{C_1} = B_{C_2} = 0$
(b) $B_{C_1} > B_{C_2}$
(c) $B_{C_1} = B_{C_2} \neq 0$
(d) $B_{C_1} < B_{C_2}$

Scanned with OKEN Scanner

13 Each of the following figures represents two concentric metallic rings in the same plane carrying electric currents. In which one of them the resultant magnetic flux density at the common center is higher? *(2nd session 23)*

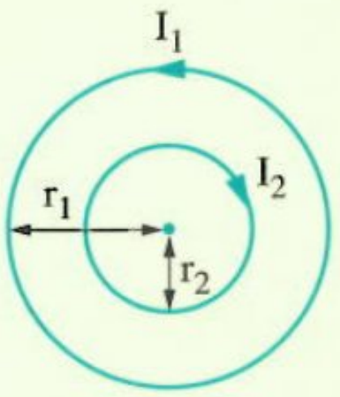


14 A circular coil has a number of turns (100 turns). A current of intensity (5 A) passes through the coil, if the radius of the coil is 2π cm, then the magnetic flux density at the center of the coil = ...

(2nd session 24)

(a) 2×10^{-3} T (b) 2 T (c) 5 T (d) 5×10^{-3} T

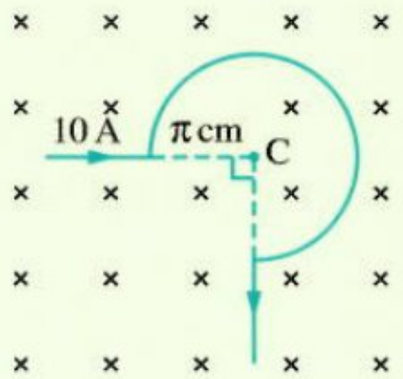
15 In the opposite diagram, two concentric circular coils lie in the plane of the page. The outer coil has a radius of 20 cm, consists of 100 turns, and carries a current of intensity 4 A in the direction shown in the figure. The inner coil has a radius of 10 cm, consists of 50 turns, and carries a current of intensity 2A in the direction shown in the figure. Then the resultant magnetic flux density at their common center equals ..... .



(Practice 25)

(a) 6.28×10^{-4} T (b) 8.42×10^{-4} T
(c) 7.36×10^{-5} T (d) 9.63×10^{-5} T

16 A straight wire is reshaped as shown in the opposite diagram. If the radius of the circular section is π cm, and the wire carries an electric current of intensity 10 A, then it is placed in an external magnetic field perpendicular to its plane, and the magnetic flux density is 1.5×10^{-4} T. Then the resultant magnetic flux density at the center (C) equals ..... . *(Practice 25)*



(a) 0 (b) 3×10^{-4} T
(c) 3.5×10^{-4} T (d) 4.5×10^{-4} T

CamScanner
Scanned with OKEN Scanner

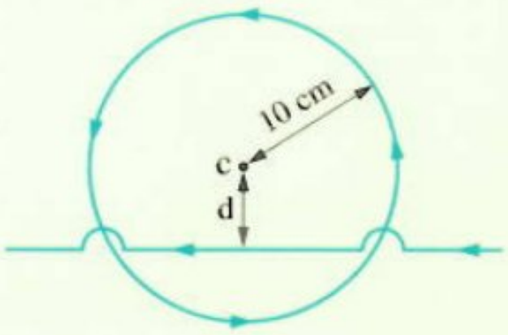
CamScanner

17 A circular coil of 50 turns carries a current of 1.4 A. Its axis is aligned with the Earth's magnetic field, which has a magnetic flux density of 5×10^{-5} T. If the coil is overturned, the total magnetic flux density at the center of the circular coil becomes double what it was initially. Then the radius of the coil is equal to ..... . *(Practice 25)*

(Knowing that: the magnetic field generated by the current in the circular coil is greater than the Earth's magnetic field, the magnetic permeability = 12.56×10^{-7} Wb/A.m)

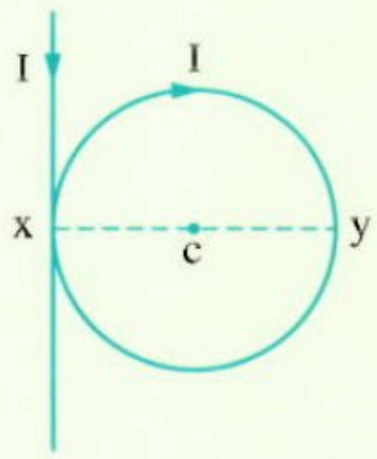
(a) 2.93 m (b) 8.8 m (c) 0.88 m (d) 0.293 m

18 In the opposite figure, an insulated circular ring carries an electric current of intensity (I), and a long insulated straight wire also carries a current of intensity (I). Both are in the same plane. If the net magnetic flux density at the center of the ring (c) is vanished, then the distance (d) equals ..... . *(Practice 25)*



(a) 3.14 cm (b) 6.28 cm (c) 3.18 cm (d) 5 cm

19 An insulated straight wire is placed tangent to a metal ring and in its plane. If an electric current of intensity (I) flows through both the wire and the ring in the directions shown in the figure, the direction of the resultant magnetic flux density at the center of the ring (c) is ..... *(1st session 25)*



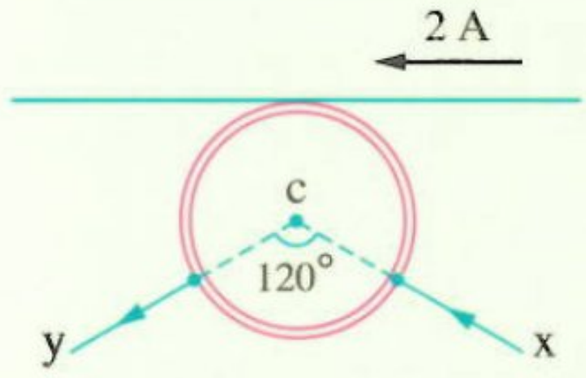
(a) perpendicular to the diameter (xy), in the plane of the ring

(b) perpendicular to the plane of the ring, inward

(c) along the diameter (xy)

(d) perpendicular to the plane of the ring, outward

20 * A metal ring has an average diameter of 10 cm and a resistance (R). An electric current of 6 A passes to it through two terminals (x) and (y). An insulated straight wire is placed in the plane of the ring and tangent to it. An electric current of 2 A passes through the wire, as shown in the figure.



What is the net magnetic flux density at the ring center (c)?

(Given that: $\mu = 4\pi \times 10^{-7}$ Wb/A.m) *(1st session 25)*

(a) 4×10^{-6} T (b) 8×10^{-6} T (c) 1.6×10^{-5} T (d) 3.2×10^{-5} T

Scanned with OKEN Scanner

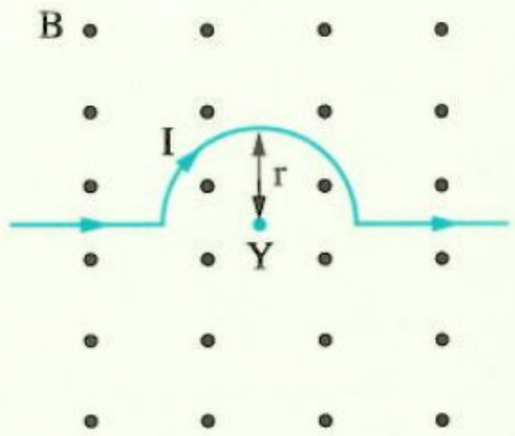
21 A circular coil of insulated wire is coiled around a soft iron core, and an electric current passes through it. The magnetic flux density at the center of the coil will be doubled when ..... .

(2nd session 25)

(a) the number of turns in the coil is halved (b) the radius of the coil is doubled

(c) the current intensity in the coil is doubled (d) the iron core is removed from the coil

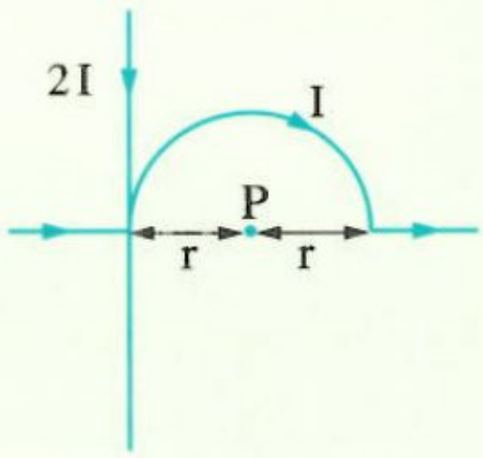
22 In the shown figure, a semi-circular loop carries an electric current of intensity (I) while being placed in a uniform magnetic field of flux density (B) that is directed perpendicularly out of the page. If the direction of the resultant magnetic flux density at point (Y) is perpendicular out of the page, which of the following relations is correct?



(Practice 26)

(a) $B < \frac{\mu I}{8 r}$ (b) $B = \frac{\mu I}{8 r}$ (c) $B < \frac{\mu I}{4 r}$ (d) $B > \frac{\mu I}{4 r}$

23 The figure shows a long, straight, insulated wire placed tangent to a metallic semi-circular loop. A direct electric current passes through each of them. The direction of the resultant magnetic field arising from the two currents at point (P) is ..... .



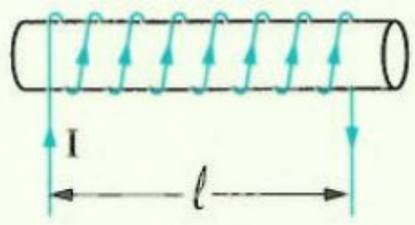
(Practice 26)

(a) perpendicular to the page, inward (b) perpendicular to the page, outward

(c) parallel to the page, to the right (d) parallel to the page, to the left

Solenoid

24 The figure illustrates a solenoid of length ℓ, cross-sectional area A and number of turns N through which a current I passes, if its turns are displaced apart so that its length gets increased to 3ℓ, the magnetic flux density along its axis ..... .



(Exp. May 21)

(a) decreases to $\frac{1}{3}$ of its initial value (b) decreases to $\frac{1}{6}$ of its initial value

(c) decreases to $\frac{1}{9}$ of its initial value (d) decreases to $\frac{1}{12}$ of its initial value

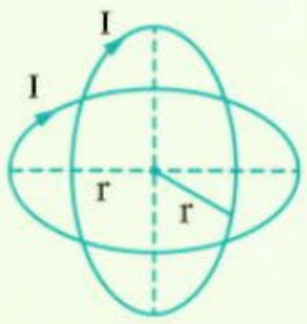
CamScanner

Scanned with OKEN Scanner

CamScanner

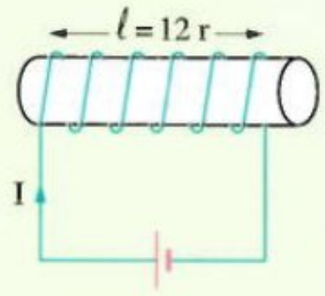
25 In the figure, a group of electric conductors, each is carrying an electric current (I).

Two perpendicular rings have the same diameter (2 r)



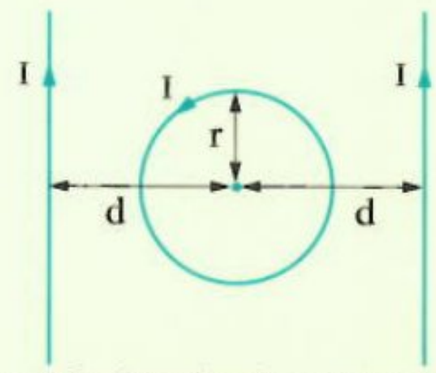
Magnetic flux density at the center of the two rings = (X)

Solenoid, its no. of turns (N = 6) and its length (ℓ = 12 r)



Magnetic flux density on the axis inside the solenoid = (Y)

Ring of radius (r)



Magnetic flux density at the center of the ring = (Z)

Which of the following mathematical relation is correct? *(1st session 22)*

(a) Z > Y (b) X = Z (c) Y < X (d) X = Y

26 A solenoid of length 20 cm consists of 100 turns and its radius is 0.1 m carries a current of 4.9 A. Knowing that the permeability of the medium inside the coil is $\frac{88}{7} \times 10^{-7}$ Wb/A.m, the magnetic flux that penetrates the face of the coil equals ..... .

(Given that: $\pi = \frac{22}{7}$) *(Exp. 23)*

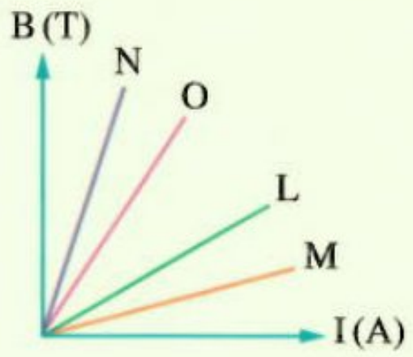
(a) 6.166×10^{-6} Wb (b) 30.8×10^{-4} Wb

(c) 6.166×10^{-3} Wb (d) 9.68×10^{-5} Wb

27 A solenoid made of insulated copper wire carries an electric current (I) and the magnetic flux density along its axis is (B), if the turns are displaced uniformly away from each other, the magnetic flux density along its axis decreases to $(\frac{B}{4})$. If we have to increase the current intensity by 3 A to return the value of magnetic flux density to its initial value (B), then the current intensity (I) is ..... . *(Exp. 23)*

(a) 1 A (b) 2 A (c) 3 A (d) 4 A

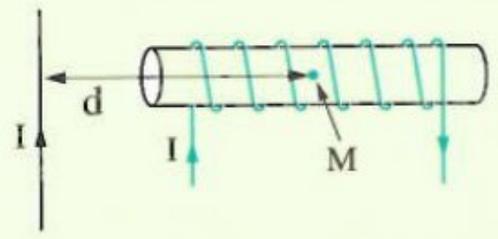
28 The graph represents the relation between the magnetic flux density (B) at the mid-point of the axis of a number of spiral coils (L, M, N and O) and the current intensity (I) passing through them, if you know that all the coils have the same number of turns and their cores have the same magnetic permeability, then the coil that has the smallest length is coil ..... . *(1st session 23)*



(a) N (b) L (c) M (d) O

Scanned with OKEN Scanner

29 The opposite figure represents a solenoid of (N) number of turns and length (ℓ), the current intensity passing through it is (I). If a long straight wire that carries a current (I) is placed in the plane of the solenoid and the direction of the electric current of the wire is perpendicular to the axis of the solenoid, the resultant of the magnetic flux density at point (M) is equal to ..... .

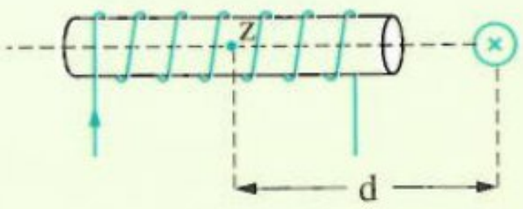


(1st session 24)

(a) $\sqrt{B^2_{solenoid} - B^2_{wire}}$ (b) $B^2_{solenoid} - B^2_{wire}$

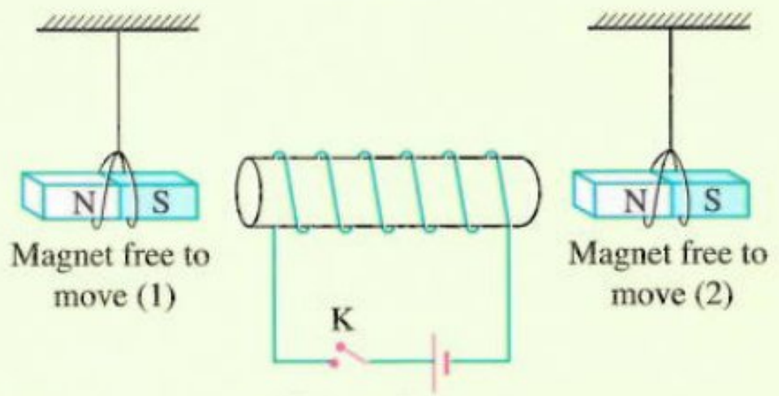
(c) $\sqrt{B^2_{wire} + B^2_{solenoid}}$ (d) $B^2_{wire} + B^2_{solenoid}$

30 The opposite figure represents a solenoid and a straight wire, each carrying an electric current. The wire is placed at a distance d from point z , which is at the midpoint of the solenoid. If the magnitudes of magnetic flux density at point z due to the current through the solenoid only and the current through the wire only are B and $\frac{4}{3}$ B respectively, and the net magnetic flux density at point z due to both the solenoid and the wire currents is B_1, then when the distance between the wire and point z increases to 2d, the net magnetic flux density at point z becomes B_2 , then ..... *(1st session 24)*



(a) $B_2 = 1.4\ B_1$ (b) $B_2 = 0.72\ B_1$ (c) $B_2 = 1.6\ B_1$ (d) $B_2 = 0.5\ B_1$

31 From the opposite figure, what happens when the switch (K) is closed? *(1st session 24)*



(a) Magnet (2) approaches the coil while magnet (1) moves away from it.

(b) Magnets (1) and (2) approach the coil.

(c) Magnet (1) approaches the coil while magnet (2) moves away from it.

(d) Magnets (1) and (2) move away from the coil.

32 A solenoid has a number of turns 14 turns and a length 22 cm carrying an electric current of 2 A, then the magnetic flux density at the mid-point of its axis ..... . *(2nd session 24)*

($\mu = \frac{88}{7} \times 10^{-7}$ T.m/A)

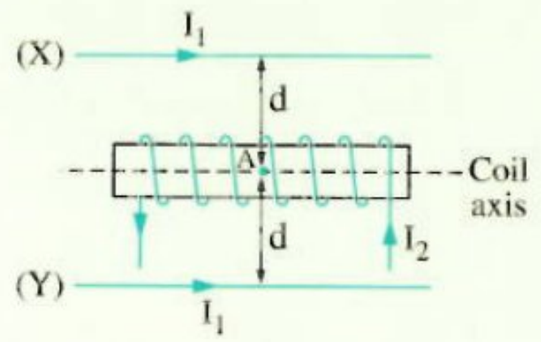
(a) 16×10^{-7} T (b) 1.6×10^{-4} T (c) 8×10^{-4} T (d) 8×10^{-7} T

CamScanner

Scanned with OKEN Scanner

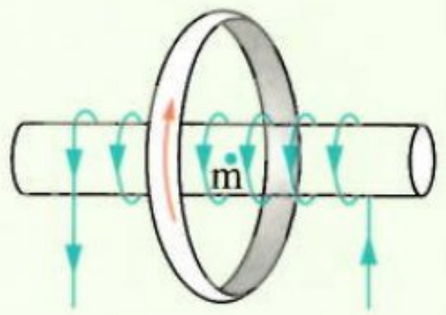
CamScanner

33 In the opposite figure, if the magnetic flux density resulted from each of wire X, wire Y and the solenoid one at a time at point A is B. Then, which of the following choices represents the resultant magnetic flux density at the same point, when the current is reversed in one of the two wires? *(2nd session 24)*



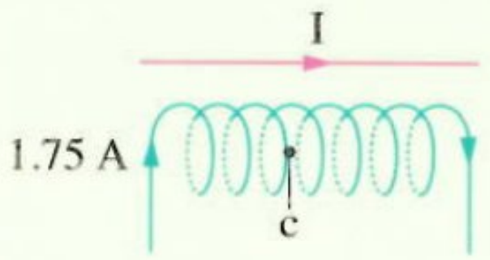
(a) $\sqrt{3}$ B (b) 5 B (c) $\sqrt{5}$ B (d) 3 B

34 A long solenoid contains 5 turns per cm of its length and a circular coil of radius $\frac{\pi}{5}$ cm and consists of 10 turns is wound around the midpoint of the solenoid and the two coils are concentric and coaxial as shown in the opposite figure, if an electric current of intensity 4 A passes through each of the two coils, the magnetic flux density at the common center (m) equals ..... . *(Practice 25)*



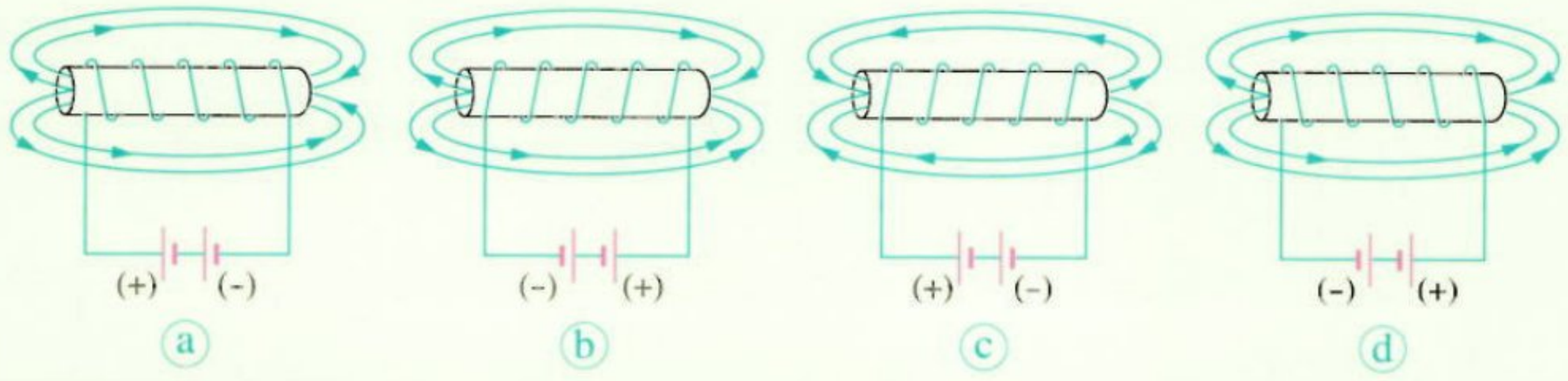
(a) 4×10^{-3} T (b) 2.5×10^{-3} T (c) 1.5×10^{-3} T (d) 6.5×10^{-3} T

35 In the opposite diagram, a solenoid contains 300 turns per meter and carries a current of intensity 1.75 A. A straight wire is placed near to the coil and parallel to the solenoid's axis. If the magnetic flux density resulting from the electric current in the straight wire at point (c) which is located at the midpoint of the solenoid's axis, is 2.33×10^{-4} T, then the net magnetic flux density at point (c) is approximately equal to ..... . *(Practice 25)*



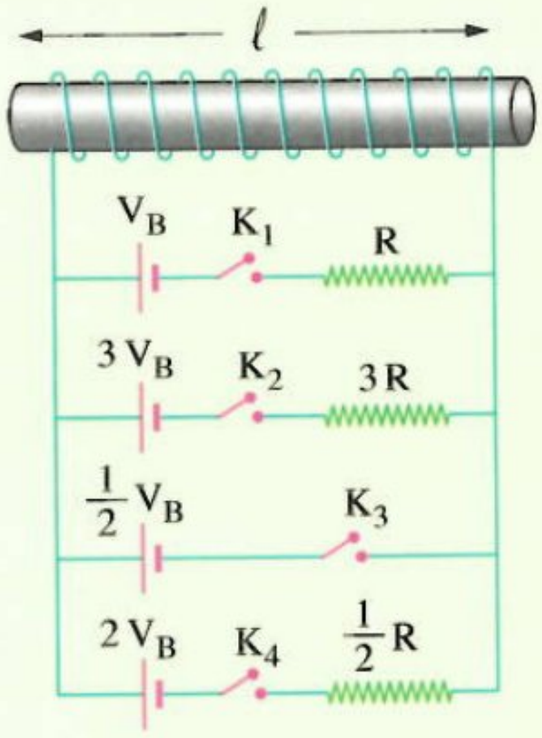
(a) 4.27×10^{-4} T (b) 6.44×10^{-4} T (c) 7×10^{-4} T (d) 8.93×10^{-4} T

36 Which of the following figures correctly describes the direction of the magnetic field generated by an electric current passing through a solenoid? *(Practice 25)*



Scanned with OKEN Scanner

37 A solenoid with a resistance of 0.5 R is connected in an electric circuit as shown in the opposite figure. The magnetic flux density at the center of the solenoid reaches its maximum value after switch ..... only is closed. *(Practice 25)*



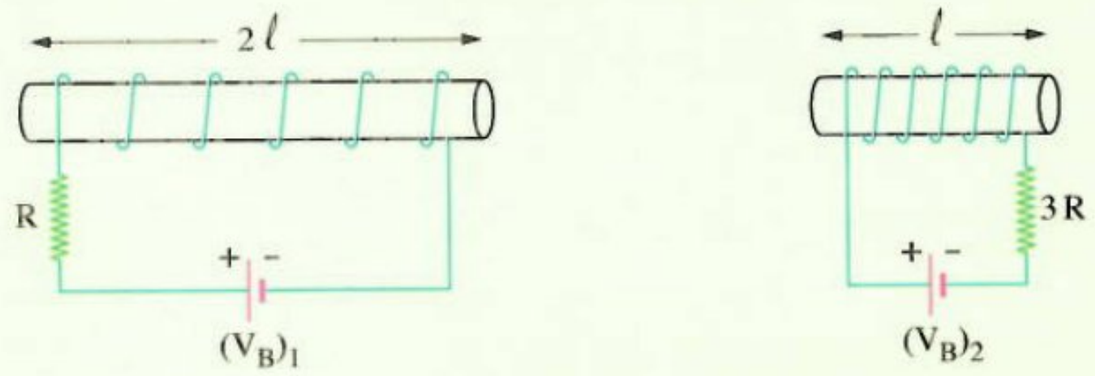
ⓐ K_1

ⓑ K_2

ⓒ K_3

ⓓ K_4

38 The following figure shows two insulated solenoids with negligible resistance, each having the same number of turns and wound around a magnetic core of the same type. Each is connected to an electric source of negligible internal resistance and an electric resistance.



Given that: the magnetic flux density at the midpoint along the axis of each solenoid is equal, and that the length of each solenoid is much greater than its diameter, then the ratio of the electromotive forces (emf) of the two sources $(\frac{V_{B1}}{V_{B2}})$ equals ..... . *(Practice 25)*

ⓐ $\frac{2}{3}$ ⓑ $\frac{1}{12}$ ⓒ $\frac{1}{4}$ ⓓ $\frac{1}{2}$

39 You have a solenoid of length (ℓ), whose turns are evenly distributed along its length. The solenoid terminals are connected to the poles of a battery having electromotive force (V_B) and negligible internal resistance. When a half of the solenoid length is removed and the remainder is connected to the same battery, which of the following physical quantities is halved? *(1st session 25)*

ⓐ The number of turns per unit length of the coil.

ⓑ The ohmic resistance of the coil.

ⓒ The current intensity through the coil.

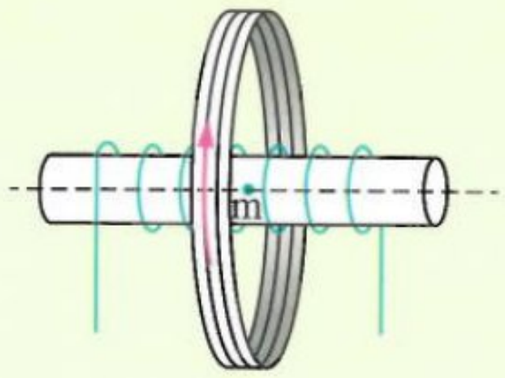
ⓓ The magnetic flux density at the center of the coil.

CamScanner

Scanned with OKEN Scanner

CamScanner

40 A circular coil of radius 10 cm has 14 turns. It is coiled around the center of an insulated solenoid consisting of 28 turns which are uniformally distributed over a length of 10 cm. The axis of the solenoid is perpendicular to the plane of the circular coil and passes through its center. A current of intensity 2 A flows through the circular coil in the direction shown in the figure.



What is the intensity and direction of the current in the solenoid to make the magnetic flux density at the center of the circular coil equal to zero? *(2nd session 25)*

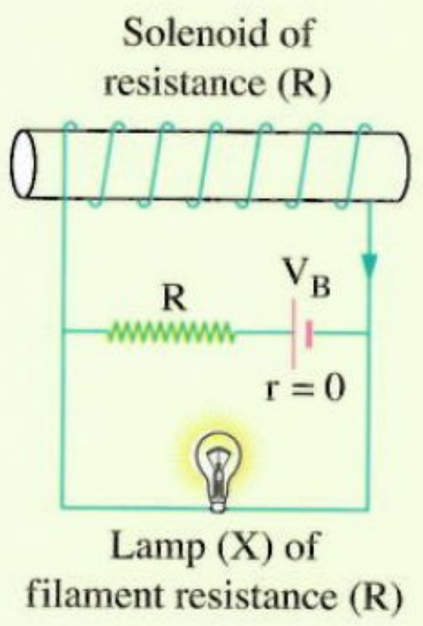
(a) 2 A, opposite to the direction of the current in the circular coil.

(b) 0.5 A, in the same direction as the current in the circular coil.

(c) 2 A, in the same direction as the current in the circular coil.

(d) 0.5 A, opposite to the direction of the current in the circular coil.

41 The figure shows a battery (V_B) of negligible internal resistance, connected to an electrical resistor (R), a solenoid of ohmic resistance (R), and an electric lamp (X) with a filament resistance (R). If the filament of lamp (X) burns out, then the magnetic flux density at the midpoint of the solenoid's length will ..... . *(Practice 26)*



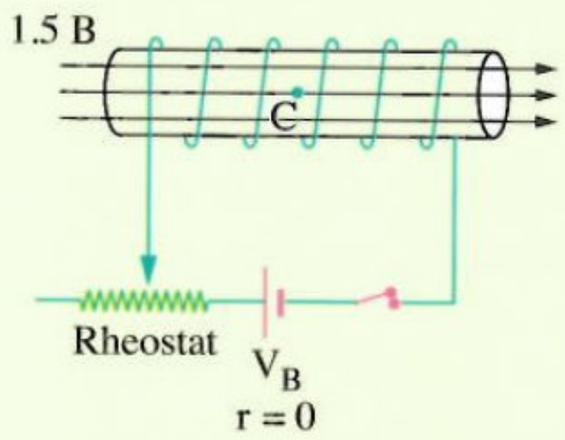
(a) decrease

(b) increase

(c) vanish

(d) remain unchanged

42 In the opposite figure: A solenoid is connected to a battery of electromotive force (V_B) and negligible internal resistance and a rheostat. An electric current passes through it, generating a magnetic field of flux density (B) at the midpoint of its axis. When the solenoid is placed in an external magnetic field of flux density (1.5B) directed to the right of the page, and the resistance taken from the rheostat is increased by one-third of its initial value, the total resultant flux density at point (C) will ..... . *(Practice 26)*



(Assuming the resistance of the solenoid's wire is negligible)

(a) increases

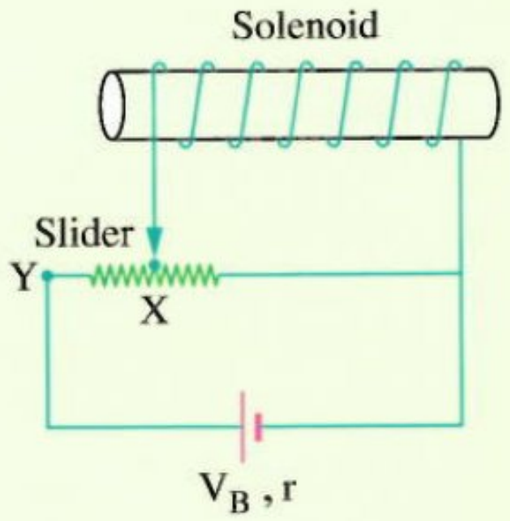
(b) decreases

(c) remains unchanged

(d) becomes zero

Scanned with OKEN Scanner

43 A solenoid, a battery of electromotive force V_B and internal resistance r and a rheostat are connected as shown in the opposite figure. When moving the slider of the rheostat from position X to position Y, the magnetic flux density at the midpoint of the solenoid's axis ..... .



(a) increases
(b) does not change
(c) decreases but does not vanish
(d) vanishes

CamScanner
Scanned with OKEN Scanner